Universitas Indonesia SDGs Summit 2026 Essay Competition

Reversing the Dominoes: Youth-Powered and Transformative Solutions for Measurable Impacts Towards a Resilient 2030 Agenda

"SIRKULA: SCALING CIRCULAR ECONOMY THROUGH DECENTRALIZED BSF BIOCONVERSION TO REDUCE ORGANIC WASTE, CREATE DECENT JOBS, AND SECURE HOUSEHOLD LIVELIHOODS IN INDONESIA"

SUBTHEME: Economic Growth

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Title of Essay : SIRKULA: SCALING CIRCULAR ECONOMY THROUGH DECENTRALIZED BSF BIOCONVERSION TO REDUCE ORGANIC WASTE, CREATE DECENT JOBS, AND SECURE HOUSEHOLD LIVELIHOODS IN INDONESIA

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(Depok, August 14, 2026)

Regards,



Leader of the Team

(David Muhammad Thorique)

I. ABSTRACT

SIRKULA connects paid organic-waste treatment with household-owned production and access to buyers. We propose a compact Sukmajaya, Depok cluster in which Pangantara households operate four GROW 10 racks, while SIRKULA coordinates sources, collection, specialist supply and settlement. A complete installation places 40 trays and handling equipment in a qualified 30 m² workspace. Entry combines Rp2,400,000 of operator capital with a proposed Rp21,600,000 equipment grant, released only after funding approval. Under the selected operating assumptions, a mature four-rack home provides Rp2,110,520 in monthly available income for 61.62 owner hours after household costs, disclosed fees and replacement saving. SIRKULA separately funds operator protection and qualified relief. Commercial treatment payments recognise household work, while direct product sales retain value with the producer. Assisted registration, practical training and linked transaction records make participation understandable and reviewable. Outsourced breeders and nurseries support launch; twelve racks and a shared hub remain separately assessed options. Expansion follows paid renewal, accepted product orders, manageable work and funded obligations. The model advances productive enterprise and resource recovery through a design that places sustained household value at the centre of growth.

II. INTRODUCTION

Global food systems generate waste at a scale that has outpaced most efforts to prevent it. In 2022 alone, the world discarded an estimated 1.05 billion tonnes of food, nearly one-fifth of everything available to consumers, with households responsible for roughly 60 percent of that total (UNEP & WRAP, 2024). Indonesia mirrors that pattern at the national level: the country generated 43 million tonnes of municipal solid waste in 2023, a 56 percent increase since 2019, and food waste alone accounted for more than 48 percent of that stream, the single largest category by a wide margin (SIPSN, as cited in University of Maryland School of Public Policy, 2025). Roughly 40 percent of everything generated nationally still receives no treatment of any kind (SIPSN, as cited in University of Maryland School of Public Policy, 2025). The question this essay

takes up is not how large that number will grow, but what economic value is lost, in the gap between a household throwing food away and a landfill receiving it.

That gap is not shrinking on its own. Indonesia's national targets called for 100 percent of municipal waste to be either reduced or properly handled by 2025, yet by 2021 actual managed coverage stood at roughly 65 percent, with reduction through reuse and recycling reaching only about 15 percent against a 24 percent target for that year (Ministry of Environment and Forestry, 2023). Methane emissions from the waste sector, driven overwhelmingly by unmanaged organic material, grew by an estimated 70 percent between 2000 and 2022 (Republic of Indonesia, 2024, as cited in University of Maryland School of Public Policy, 2025). The direction of travel is consistent across every data source available: organic waste is growing, national handling targets are being missed, and the emissions consequence is compounding.

Indonesia already runs an economy built to recover value from waste, and it is a large one. An estimated two million people work as pemulung, informal waste pickers who sort and resell recyclable material for a living (International Alliance of Waste Pickers, as cited in No-Trash Triangle Initiative, 2024), though this figure comes from advocacy-sector estimation rather than a national census and should be read as indicative rather than precise. What that economy is built to recover, however, is almost entirely the inorganic fraction: plastic, metal, glass, paper, the materials with a standing resale price. Individual waste pickers in a five-city Indonesian study collected an average of 43.87 kilograms of material a day, achieving a recycling rate of roughly 12 percent, for a monthly income of about US$91.70, well under the legal minimum wage (Kristanto, Kemala, & Nandhita, 2022). In Depok and Bekasi specifically, waste pickers reportedly earn only about one-third of the regional minimum wage (Modern Diplomacy, 2025). Organic waste, the single largest fraction of everything discarded, sits almost entirely outside this livelihood system, not because it lacks value, but because nothing currently prices it as an input the way a scrap dealer prices a kilogram of plastic.

The technology needed to price that value has existed for well over a decade. A field study in Costa Rica established as early as 2011 that black soldier

fly larvae could digest mixed municipal organic waste at scale, reducing it by 65.5 to 78.9 percent while producing a prepupae by-product explicitly identified as opening "new economic niches for small entrepreneurs in developing countries" (Diener, Studt Solano, Roa Gutiérrez, Zurbrügg, & Tockner, 2011).

That mass-conversion logic has since been replicated on Indonesian ground. A case study conducted in Tasikmalaya City documented an average yield of 8.33 kilograms of black soldier fly larvae from every 25 kilograms of organic feedstock processed, alongside frass and larval-skin by-products with measurable organic carbon content suitable for further valorization (Jupesta & Permana, 2025). The biology, in other words, is not the open question. What a kilogram of household food scraps can become, and roughly how much of it, has already been measured, published, and reproduced across different cities and research teams.

What these two decades of documentation share is a consistent finding pitched at the wrong scale for the problem they describe. Every study cited above measures conversion, not livelihood: kilograms of larvae per kilogram of feedstock, percentage waste reduction, organic carbon content of a by-product. None of them price the material at intake, protect the income of the person doing the conversion, or specify how many households a single breeding operation can reliably supply. The same imbalance shows up on the human side: the waste pickers already documented above are paid for what they can resell, not for what they prevent from reaching a landfill, and nothing in Indonesia's current informal waste economy compensates a household for accepting and processing organic residue the way it compensates a picker for a kilogram of PET bottles.

None of this points to a missing technology. SIRKULA is the model built to close it. Rather than proposing another isolated bioconversion demonstration, SIRKULA organizes trained home-based operators, called Pangantara, into intake, grow-out, and harvest enterprises within their own qualified home service areas, supported by a shared breeding and nursery hub that supplies starter biology without requiring every household to run its own colony. Livelihood-Secured Scaling is the organizing principle behind that structure: new Pangantara nodes are only added where recurring suitable feedstock, protected operator net income, safe processing capacity, and credible buyer demand can all be demonstrated

together, so that expansion does not come at the cost of an existing partner's earnings. In direct alignment with SDG 8's mandate for decent work and inclusive growth, SIRKULA is deliberately anchored in a single, concrete first test: a pilot in Depok, the same city where informal waste pickers today already earn a fraction of the minimum wage from the materials that already have a market. What the Depok pilot has to prove is narrow by design, because what it proves there is the basis for arguing the model can work anywhere else in urban Indonesia.

III. IN-DEPTH ANALYSIS

The Scale and Cost of an Underused Resource

Indonesia generated between 23 and 48 million tons of food loss and waste annually between 2000 and 2019, equivalent to 115–184 kilograms per capita per year, according to a joint study by the Ministry of National Development Planning (Bappenas), the World Resources Institute, Waste4Change, and the UK Foreign, Commonwealth and Development Office (Bappenas et al., 2021). Organic material dominates the national waste composition, and Indonesia is frequently cited as the world's second largest producer of food waste after Saudi Arabia (Waluyo & Kharisma, 2023). This is not merely an environmental statistic; decomposing organic waste in open landfill conditions releases methane, a greenhouse gas with a substantially higher short-term warming potential than carbon dioxide, meaning that every ton left unmanaged carries a compounding climate cost alongside its lost material value.

The economic magnitude of this loss is equally significant. Bappenas et al. (2021) estimated that food loss and waste cost Indonesia between Rp213 trillion and Rp551 trillion per year, equivalent to roughly 4–5% of national GDP. This figure captures only the value of food itself; it does not account for the secondary economic value that the residual organic mass could generate if diverted into a productive conversion pathway before disposal. The gap between what is currently lost and what could be recovered is precisely the space in which a bioconversion-based enterprise model becomes economically legible.

Technical Feasibility of Bioconversion at Household Scale

Black soldier fly (Hermetia illucens) larvae are among the few biological agents capable of converting heterogeneous organic waste into a stable, marketable output without requiring industrial infrastructure. Indonesian evidence for this already exists at small scale: a study conducted at the Faculty of Engineering cafeteria, Universitas Indonesia, found that 87% of the cafeteria's 46.27 kg of daily organic waste was suitable bioconversion substrate, and that a 12-day bioconversion cycle reduced waste mass by 75%, with a Waste Reduction Index of 6.25% per day and larvae biomass containing 29.1% protein and 11.86% fat (Ibadurrohman et al., 2020). This result matters analytically because it demonstrates that meaningful, quantifiable conversion outcomes are achievable at a scale comparable to a single household or small institutional kitchen. It directly supports the premise that a home-based, distributed processing model is technically defensible rather than merely aspirational.

At the same time, bioconversion performance is sensitive to feedstock quality, moisture content, and temperature control, and that only low-cellulose organic fractions are suitable substrate (Ibadurrohman et al., 2020). This is an important qualifying detail: it means household-level conversion cannot be treated as a passive drop-off process. It requires an inspection and sorting step between intake and grow-out, which has direct implications for how a distributed enterprise model must be designed if it is to protect both biological performance and the economic value of the resulting larvae.

Demand-Side Rationale: An Import-Dependent Protein Market

The output side of this equation is equally consequential. Indonesia's animal feed industry remains heavily dependent on imported protein: the poultry sector alone imported approximately 4.4 million tons of soybean meal to meet feed demand, according to USDA data cited in a recent review of Indonesian poultry sustainability (Sumiati et al., 2025). This dependency exposes domestic feed prices to international commodity volatility and represents a structural vulnerability in the broader livestock value chain. Black soldier fly larvae meal, as

a locally producible, high-protein feed ingredient, sits directly against this import exposure, offering a plausible substitution pathway rather than a speculative one.

This demand, however, should not be overstated. Controlled trials on broiler chickens found that replacing 15% of soybean meal crude protein with defatted Hermetia illucens larvae meal did not impair growth performance or digestibility, but replacing 30% did (Hartinger et al., 2021). This finding matters for realistic market sizing: it indicates that BSF-derived protein currently functions as a partial, blended substitute within existing feed formulations rather than a wholesale replacement for imported meal. Any economic projection for a BSF-based enterprise should therefore treat the addressable feed market as bounded by this substitution ceiling rather than by the full volume of Indonesia's soybean meal imports.

Root Cause I: Structural Weakness of the Informal Waste Value Chain

Even where recyclable material already has a recognized resale market, the actors who physically handle it capture the least value. A pressure-state-response analysis of Balikpapan's informal waste sector found that scrap collectors receive the smallest share of profit within the recycling chain, alongside significant volumes of collected material going unsold or unmanaged by intermediaries (Harfadli et al., 2024). If this distributional pattern holds even for materials with an established secondary market such as plastic, it follows that organic waste, which has historically had no comparable resale market at all, is structurally excluded from any value capture whatsoever under the conventional informal recycling system. The absence of an organic-waste equivalent to the scrap trade is not a gap that existing informal infrastructure will close on its own; it requires a purpose-built mechanism that assigns cash value to material the current system treats as worthless.

Root Cause II: Uneven Income Capture Even Within Existing Community Enterprises

Indonesia's most established community-based waste enterprise model, the bank sampah (waste bank), demonstrates that structured, cash-for-waste mechanisms can generate real household income. However, evidence from a

waste entrepreneurship training program in Semarang shows that this income benefit is not evenly distributed: participants who became waste bank managers experienced substantially greater income gains than informal waste collectors within the same program, a disparity the authors attribute to unequal market access and structural inequality within the solid waste management value chain (Astuti et al., 2026). This finding is directly instructive for any organic-waste enterprise seeking to scale: simply replicating a cash-for-material intake point, without deliberately protecting the income position of the frontline operator, risks reproducing the same hierarchy in which value concentrates upstream of the person doing the physical labor.

Root Cause III: The Regulatory and Institutional Gap

At the policy level, Indonesia's circular economy framework remains comparatively underdeveloped. A comparative legal-policy analysis found that, unlike Germany, the Netherlands, and China, Indonesia lacks a dedicated circular economy act capable of providing binding guidance for waste and food-waste management across government, private sector, and community actors (Waluyo & Kharisma, 2023). This institutional gap means that any organic-waste enterprise operating in Indonesia today does so without a strong formal regulatory scaffold to rely on for waste sourcing, quality standards, or buyer protection. In practice, this shifts the burden of establishing trust, quality assurance, and fair pricing onto the enterprise's own internal design rather than onto external regulation.

Convergence Toward a Livelihood-Anchored Design Requirement: A Synthesis

All of the three root causes reveal a pattern that is structural rather than incidental. The technical capacity to convert organic waste into valuable protein exists and has been demonstrated at small scale within Indonesia (Ibadurrohman et al., 2020); a bounded but genuine market demand exists on the output side (Sumiati et al., 2025; Hartinger et al., 2021); yet the intermediate stage, where material is collected, sorted, and converted, is precisely where existing systems consistently fail to protect the income of the people doing that work (Harfadli et al., 2024; Astuti et al., 2026), compounded by the absence of a regulatory

framework that might otherwise correct this imbalance (Waluyo & Kharisma, 2023). What is required, therefore, is not another collection mechanism or another biological conversion method in isolation, but a model that treats fair, protected household income as a binding design constraint from the outset rather than as a hoped-for downstream outcome.

IV. RECOMMENDATION

IV.1. A complete household enterprise

Our proposed starting point is a complete four-rack Pangantara installation: four GROW 10 units, 40 removable trays, a receiving and weighing station, guarded preparation equipment, a harvest station, washing facilities, curing bins and finished-stock storage. The installation fits an existing covered 30 m² workspace, separated from living areas and food preparation. This arrangement gives the operator a practical sequence from incoming material to saleable output, with enough space for cleaning and finishing rather than treating the racks alone as an enterprise. Its layout, equipment allowance and procurement specifications are set out in Appendix 3.1.

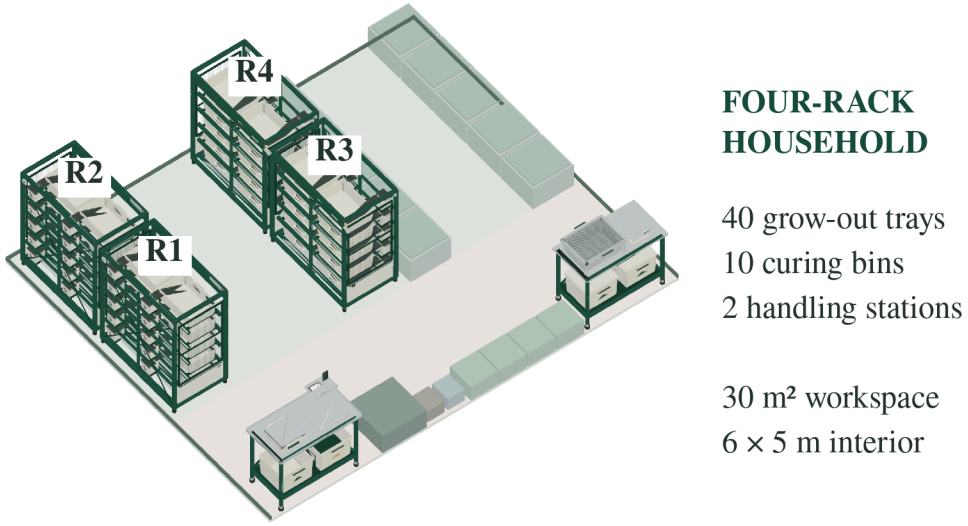


Figure 1. Complete four-rack Pangantara installation.

SIRKULA brings together five participants around that installation. Residents supply separated eligible food scraps; commercial customers purchase

scheduled treatment; Pangantara performs grow-out and owns its equipment and output; specialist suppliers provide starters and technical support; buyers accept identified product lots. SIRKULA coordinates these relationships and keeps capacity, service and settlement aligned. Residents gain a paid route for suitable material, customers gain documented treatment, and the household gains both remuneration for handling work and a market for its products. The participant value proposition explains these benefits alongside each party's obligations (Appendix 6.1).

We begin in a compact Sukmajaya catchment where proximity can make daily service practical. Sukmajaya's 253,712 residents and density of 14,606 people/km² in 2025 support this geographical choice, while actual service volume comes from eligible sources within the route (Badan Pusat Statistik Kota Depok, 2026). The commercial plan reserves capacity; the operator accepts only suitable delivered material. This links the local waste problem to a defined service and an owned productive asset. The broader value and strategy analysis connects that choice to the proposed enterprise's development priorities (Appendix 6).

The Rp24,000,000 package at 2027 planning prices combines a Rp2,400,000 household contribution and a proposed conditional equipment grant of Rp21,600,000. An illustrative Rp361,146 household working-cash buffer covers the pre-receipt interval. A signed handover identifies the household owner, safe-use duties and exit terms; normal trading failure does not automatically turn the grant into household debt. The funder bears grant-capital risk, while the operator's own capital and variable earnings remain exposed. The complete household account and funding terms preserve these different responsibilities (Appendix 8.1).

Prospective CSR/TJSL and philanthropic providers are funding partners to approach, rather than committed sponsors. Procurement follows an actual allocation, a qualified site and an itemised offer accepted by the operator. Applicants can use assisted community registration, so digital literacy does not decide who can enter. Site, time, consent and working-cash checks make the offer concrete before commitment; practical training and a commissioned cycle then establish the operating routine (Appendix 7.1). The registration and

implementation framework carries these decisions through to continuing service (Appendix 7).

IV.2. How material and payment move

A resident separates suitable food scraps or produce residues and presents them at agreed intake. A commercial customer prepares the separated stream for scheduled collection. At RECEIVE, the operator checks source and condition, records gross weight, tare and rejects, then accepts only the eligible mass that fits available capacity. Plastic, hazardous residues and unsuitable material leave the productive stream through an identified destination. GROW 10 trays then support staggered grow-out, and HARVEST separates larvae from residue. Equipment views show how these three household functions connect (Appendix 3.2).

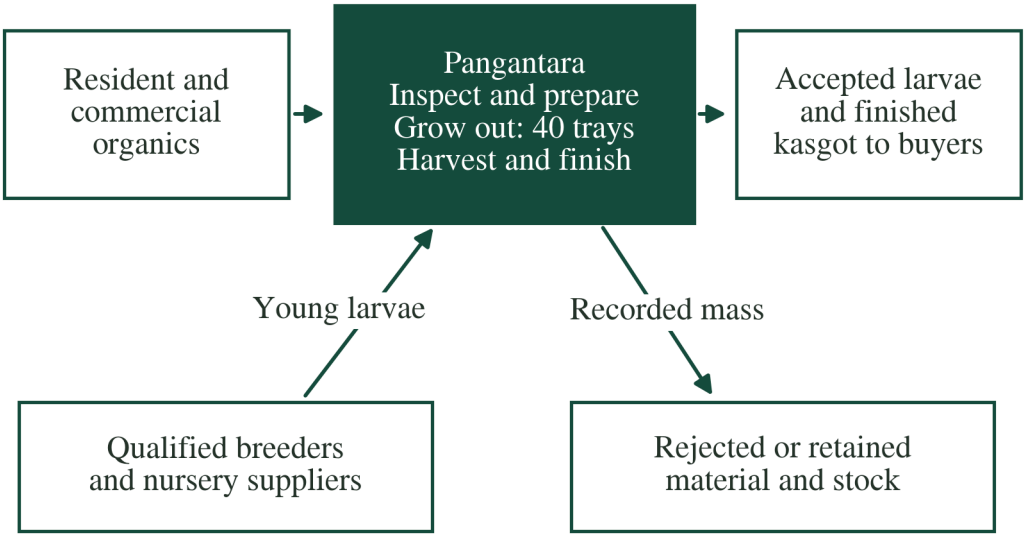


Figure 2. Organic material and household-owned outputs.

The payment path recognises two different source relationships. Residents receive Rp400 per accepted kilogram, funded by SIRKULA. Commercial customers pay SIRKULA Rp1,500,000 per month for a 200 kg allowance, with Rp5,000 charged per accepted excess kilogram under an active agreement. SIRKULA pays the household Rp3,700 per accepted commercial kilogram for handling and treatment. A reservation therefore provides service capacity, while

recorded accepted weight determines treatment remuneration. Product buyers pay Pangantara directly, and SIRKULA invoices a disclosed 10% sales-service fee. Ownership, custody and settlement rules identify each handoff and payer (Appendix 2.1).

This separation makes the household's work visible before products are sold. A customer invoice, a handling amount earned and a payment received are three different records; an unpaid invoice remains a receivable. Household sales, coordinator fees and resident rewards are also distinct flows. Internal transfers cancel when the whole network is consolidated, so they cannot become additional external revenue. The operating model uses that discipline to protect both price transparency and the meaning of household earnings (Appendix 2).

Qualified breeders and nurseries supply young larvae to the household calendar. Forty trays each receive 15 kg across an 18-day occupation period, including harvest and turnaround, giving nominal average capacity of 33.33 kg/day. At 85% utilisation, the mature account accepts 862.40 kg in an average month. The source mix assigns 80% to commercial material and 20% to residents. Staggered tray starts spread feeding and harvest across days, while recorded cumulative feed prevents a busy route from exceeding the biological work space (Appendix 4.1).

Harvest is the start of product preparation rather than the end of the material account. Fresh larvae are graded for buyer acceptance; residue undergoes controlled finishing and screening before kasgot release. Retained biomass, rejects, unsold stock and captured liquid remain visible. The starting model assigns no revenue to raw liquid. Drying or media blending would divert a declared portion of existing output and incur processing costs, so the same harvest cannot be sold twice. The production and work account connects input capacity with these output and storage requirements (Appendix 4).

IV.3. Work, equipment and two distinct scales

Four racks are designed as supplementary household enterprise. The complete normal owner-work account totals 61.62 hours/month, covering checks and records, sorting, preparation, feeding, tray work, harvest, cleaning, finishing,

packing and dispatch. Powered chopping helps preparation, but leaves the surrounding tasks intact. Staggered finishing targets a high-load day near 2.4 hours. Curing-bin capacity and stock-age limits matter because work and material continue after larvae leave the trays; the workload and continuity account includes those tasks (Appendix 4.2).

BREED B01 represents specialist breeding, while BIBIT N01 covers incubation and early larval handling. Purchasing these functions at launch allows the household to focus on grow-out without maintaining its own adult colony or nursery. The supplied specialist concepts show how a later shared facility could support that work (Appendix 3.3). Together, the household stations and specialist facilities divide receiving, grow-out, harvest and starter supply into practical functions with separate workspace and operating requirements (Appendix 3).

Twelve racks form a separate larger enterprise: 120 trays, 78 m² and a Rp62,500,000 package. Its mature static comparison gives Rp6,547,043 of monthly available income for 146.31 owner hours, or Rp44,748 per active hour. That workload belongs to a different household decision from the four-rack supplementary activity. Only four-rack homes enter the selected monthly rollout; the larger case does not inherit its forecast or historical scenario coverage. The financial comparison therefore keeps the two investment choices distinct (Appendix 8).

Operators may remain at four racks. Adding equipment to the same home requires fresh space, workload, capital and buyer checks, including the extra finishing and stock that larger output creates. Similarly, a shared hub needs its own occupancy, staffing, maintenance, funding and supply-capacity decision. The current full-cost comparison favours outsourced support within modeled capacity; the coordinator cost account explains why ownership of specialist infrastructure is a later option (Appendix 8.2).

IV.4. Markets and household value

SIRKULA sells a treatment service and household products into separate markets. For treatment, the Depok establishment registry provides a broad starting universe; eligibility, route geography and paid acceptance narrow it. We screen

30–50 addresses in the proposed catchment and seek 10–20 eligible prospects as outreach targets. One commissioning home can reserve one 200 kg plan, rising to three at mature capacity. These capacity and geography filters give the treatment-service TAM, SAM and SOM a practical role in selecting the first route (Appendix 5.2).

For products, prospective fresh-larvae outlets include compatible aquaculture and feed buyers, while finished kasgot targets qualified growers and media retailers. Buyer specifications, reachable delivery conditions and accepted repeat orders determine the obtainable market. At mature assumptions, one home sells 86.71 kg of fresh larvae and 135.41 kg of kasgot per month. The product-outlet analysis keeps these modeled sales separate from sector statistics and buyer demand (Appendix 5.3). The complete market appendix connects both demand streams to product preparation and commercial terms (Appendix 5).

The starting prices are Rp4,000/kg for fresh larvae, using the December 2024 Depok reference, and an assumed Rp1,500/kg for kasgot (Tohari, 2024). Fresh sales preserve household ownership until the buyer accepts the lot; a raw-material sale then transfers title. Under toll processing, the household instead retains output ownership and pays the processor's agreed charge. Grade, moisture basis, recovery, rejects and payment terms must accompany either route. The product and sales framework supplies those terms and the evidence required for optional processing (Appendix 5.1).

Treatment prices must be assessed against comparable service scope. Depok lists Rp50,000/m³ for restaurants and cafés and Rp750,000/month for commercial container transport (Pemerintah Kota Depok, 2025). Waste4Change residential collection starts at Rp30,000/home/month for 1–2 weekly pickups, subject to a 100-home cluster and one-year term (Waste4Change, n.d.-c). Those units and frequencies differ from SIRKULA's daily commercial plan. The tariff comparison preserves them rather than manufacturing a per-kilogram equivalence, and directs validation towards matched quotations, paid service and renewal (Appendix 5.4).

The household financial story begins with Rp3,102,654 of mature monthly receipts: Rp2,552,692 from commercial treatment work and Rp549,963 from

product sales. After selling charges and household cash costs, Rp2,402,265 remains. Earmarking Rp291,745 for equipment replacement leaves Rp2,110,520 available to the operator. The allocation in Figure 3 shows how paid treatment work anchors earnings and how renewal saving protects the asset that produces them. The selected household account preserves each deduction and payer rather than using product turnover as take-home income (Appendix 8.1).

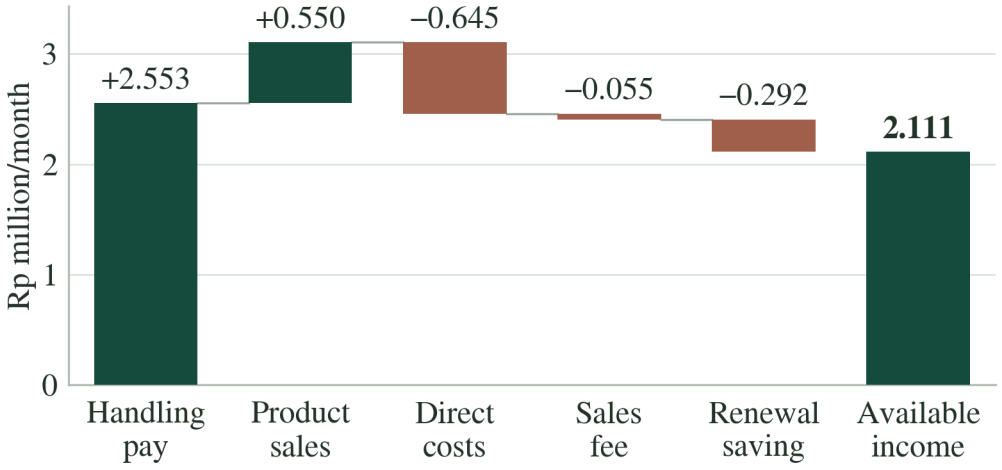


Figure 3. Allocation of mature four-rack household receipts.

Source: Selected household financial model, 2027 planning prices.

Available income is Rp34,250 per owner hour. Valuing time at Rp25,000/hour and existing workspace at Rp375,000/month leaves an economic residual of Rp194,999. This second test asks whether the activity rewards the household resources it uses, beyond generating positive cash. Under unchanged prices and source mix, its viable delivered-volume floor is 753.46 kg/month, including 602.77 commercial kg. Its economic work ceiling is 69.42 hours/month, which binds before the 75-hour supplementary-work limit. Alternative terms and headroom tests show where adjustments would change these results (Appendix 8.4).

The coordinator must also sustain collection, specialist supply, paid administration and protection. The selected forecast first covers annual running costs in 2028; by 2036, revenue covers 119.88% of those costs. Required external

funding peaks at Rp416,335,754, including the operating reserve, with Rp123,950,910 required in the first 24 months. Committed financing remains zero. Detailed annual cash, equipment support and liquidity schedules distinguish a viable service margin from the capital needed to deliver it (Appendix 8.3).

The strategic assessment turns these dependencies into implementation priorities: paid route renewal, accepted products, timed complete-cycle work, funded equipment and dependable support. The author-assessed IFE and EFE response scores remain 1.80 and 1.82 under the stated 1–4 method (David et al., 2020). Their purpose is to order the work, with price acceptance and process performance addressed before new homes are added (Appendix 6.2).

Sensitivity is useful because a household can lose value even when reserved-plan receipts protect coordinator revenue. We retain the historical scenario archive as supporting analysis (Appendix 9). Its declared design records 10,000 assumption-driven draws that preceded later billing, capacity and protection revisions (Appendix 9.1). Captured outcomes, rank associations and numerical stability therefore inform the questions to test, while current decisions use the selected cash account and deterministic operating alternatives (Appendix 9.2).

IV.5. Commissioning and paid partnerships

Preparation starts with seven consecutive days of source measurement and discussions with prospective buyers. Eligible mass, rejects, collection windows and current charges establish the route; product grade, quantity and payment terms establish the outlet. Site approval, funding and a priced service agreement then precede procurement. Training follows the whole work sequence, including guarded preparation, abnormal batches, finishing, cleaning and a reconciled settlement. The proposed schedule and witnessed competencies make these activities reviewable (Appendix 7.2).

Each launch relationship has a defined accountable function. The technical lead agrees starter age, delivery, quality, backup and remedies with breeders and advisers. Operations agrees stops, windows and diversion with the route contractor; the home seller and sales coordinator agree product acceptance and

settlement with buyers. Finance approves grants and prepares statements, while an operator reviewer handles contested charges independently. Qualified people may combine functions, with their paid work still included in the budget. Partnership terms and approvals assign these duties without implying separate unpaid staff (Appendix 7.3).

Table 1. Preparation, commissioning and review schedule.

Stage	Relative timing	Completion evidence
Measure sources and discuss service	Preparation weeks 1–3	Seven-day material record, priced route and buyer terms.
Approve site and capital	Preparation weeks 2–4	Qualified workspace and signed funded allocation.
Install and train	Preparation weeks 4–6	Safe complete kit, competencies and starter calendar.
Test paid service	Operating month 1	Four paid weeks: weights, work, payments and renewal.
Qualify production	Operating months 1–3	Three grow-out cycles, two paid larvae lots and a qualified kasgot sale.
Review cohorts and additions	Months 3, 6 and 12	Household value, accepted demand, support and funded obligations.

The first four paid weeks assess whether promised service reaches customers and money reaches the right participant. Three operating months add complete grow-out evidence and qualified product sales. Subsequent months 6 and 12 reviews examine continuity, retention, real income and reserve adequacy. Added homes follow both recurring demand and finance capacity: a successful first harvest alone does not release another equipment commitment. This sequence keeps expansion tied to the household enterprise’s continuing performance rather than enrolment totals.

IV.6. Digital coordination, quality and household rights

The SIRKULA platform links sources, accepted and rejected weights, household capacity, batch dates, product lots and settlement. Residents can inspect reward status; commercial customers can follow service and invoices; Pangantara can check intake, stock and earnings; coordinators can reconcile routes and exceptions. Figure 4 shows the prototype intake view distinguishing accepted weight from rejected material. These demonstration records illustrate the interface; operating use requires authenticated permissions, correction history, backups and reconciled restored records (Appendix 2.2).

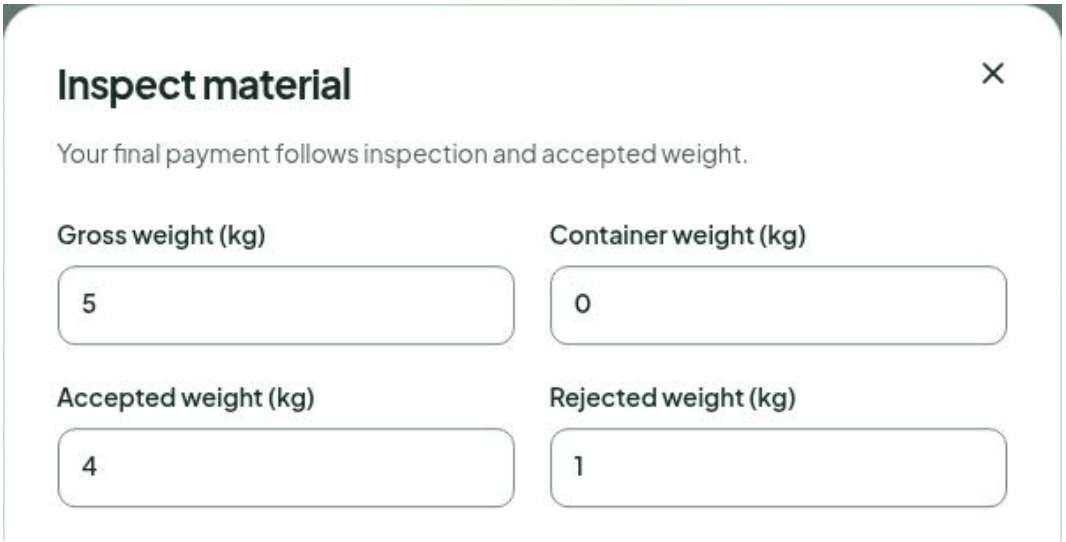


Figure 4. Historical prototype intake record (demonstration data).

Source: SIRKULA prototype.

The website and financial workbook access panel appears first among the appendices (Appendix 1). Its briefly labelled QR codes and source links provide direct access to both resources (Appendix 1.1). In daily use, assisted entry and witnessed weighing remain available alongside the platform, and original records survive corrections. This gives households a route to challenge unexplained charges and lets the coordinator trace a product or payment back to its source.

Quality control follows the material through receiving, growth, finishing and dispatch. Source age and condition continue through delivery; an unsafe load is segregated rather than diluted with clean feed. Abnormal trays are isolated, and larvae are graded before sale. Kasgot release requires maturity, hygiene, moisture

and intended-use evidence; raw frass is not automatically finished fertilizer. Lot records preserve allocation and downstream destination so implicated stock can be held or recalled (Appendix 10.2).

SIRKULA budgets Rp128,800 per active home/month for operator JKK and JKM contributions and two hours of qualified standby relief. Household PPE and hygiene remain within the existing operating allowance, and incident review uses the allocated professional-services budget. Coverage, worker classification and actual rates are confirmed before intake. The quality and risk framework ties these paid safeguards to response owners (Appendix 10).

Underfill calls for consolidation of eligible contracted loads within spare capacity, renewal work and a revised route; expansion pauses if operator income or economic residual fails. Overload calls for rescheduling finishing, qualified relief within its budget or reduced starts. Contamination, nuisance or abnormal biology triggers isolation and technical approval before restart. Further help needs funded approval, so a control does not silently become free household labour. The risk tables connect each trigger, action, owner and cost (Appendix 10.1).

Household ownership also needs a voice in the relationship. Operators review fees and expansion through representation, receive advance notice of price changes and can seek independent review of disputed statements. Voluntary exit reconciles equipment title, stock, reserves, receivables and obligations. These rights make the same transparency that coordinates production useful to the person whose home and time support it.

IV.7. Contribution and livelihood-secured growth

SIRKULA's strongest SDG link is target 8.3 through productive household enterprise, with target 9.3 supported by buyer and funding access. Documented equipment ownership contributes to the asset-access dimension of 1.4; paid protection, safer routines and incident controls support 8.8. We measure available income, all owner and relief hours, payment punctuality, participation and safety outcomes. The target-level framework defines these contributions and their denominators, rather than equating enrolment with decent work (Appendix 11.1).

A mature four-rack home treats about 10.35 tonnes of eligible material annually. The selected demand-constrained sequence reaches 19 closing homes and 196.15 tonnes of accepted feed in 2036, with commissioning timing retained in the annual account. The direct environmental contribution is resource recovery under target 12.5. Food-waste prevention under 12.3 would require a separate generated-waste baseline. Carbon receipts remain zero. The impact appendix joins material recovery with household and service outcomes (Appendix 11).

Livelihood-secured growth means adding a home only when paying sources, accepted repeat orders, safe capacity and funded support preserve operator value. Reviews retain active, suspended and exited homes in the original cohort denominator, report income and workload distributions, and compare real earnings using a disclosed price basis. Replication repeats the local source, buyer, route, workspace and financing assessment. Another community inherits these principles, while developing its own service case; the review and replication rules specify that decision (Appendix 11.2).

V. CONCLUSION

SIRKULA gives organic-waste treatment an accountable household enterprise: owned equipment, remuneration for accepted commercial material, direct product sales and transparent settlement. The four-rack starting design connects a complete installation with outsourced specialist supply and funded operator safeguards. Under the selected assumptions, it leaves Rp2,110,520 in monthly available income for 61.62 owner hours after costs and replacement saving. Its practical strength lies in recognising both treatment work and product value, while preserving resources for continued production. Twelve racks and a shared hub remain separate investment choices. We propose commissioning the compact Sukmajaya cluster through paid service, qualified products and complete-cycle work, then expanding only where household value and funded obligations endure. That sequence turns resource recovery into a route towards productive enterprise, market access and safer work whose success can be measured in the lives and accounts of its operators.

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APPENDICES

Supporting evidence is organised by operating decisions, household economics and implementation. Select a heading to open its subsection.

Appendix 1 Website and workbook access

1.1 SIRKULA website and financial workbook

Appendix 2 Operating model and digital coordination

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2.2 Linked records and platform prototype

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3.1 Household installation equipment and storage

3.2 Grow out receiving and harvest equipment views

3.3 Specialist breeding and nursery equipment views

Appendix 4 Capacity material balance and operator work

4.1 Tray calendar and mature physical account

4.2 Complete work account protection and continuity

Appendix 5 Products service markets and tariffs

5.1 Products processing and direct household sales

5.2 Treatment service market size

5.3 Product outlet market size

5.4 Service tariffs and market implementation

Appendix 6 Participant value and strategic priorities

6.1 Participant value

6.2 Strategy and planned capabilities

Appendix 7 Registration commissioning and partnerships

7.1 Registration and commissioning pathway

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7.3 Partnership terms responsibility and approval

Appendix 8 Household economics coordinator costs and funding

8.1 Household account capital and distinct enterprise options

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Appendix 9 Historical scenario analysis

9.1 Historical scenario design and provenance

9.2 Captured distributions association and stability

Appendix 10 Safeguards operational risks and product release

10.1 Funded safeguards exceptions and risk owners

10.2 Product lot evidence and release

Appendix 11 Impact measurement review and replication

11.1 Target contributions outcome definitions and denominators

11.2 Review stages and replication decisions

Appendix 1 Website and workbook access

1.1 SIRKULA website and financial workbook

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SIRKULA Website	Sustainability Model
https://tinyurl.com/SirkulaPrototype	https://tinyurl.com/SirkulaSustainabilityModel

Appendix 2 Operating model and digital coordination

2.1 Ownership handoffs and service settlement

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Pangantara owns the kit and BSF output. Acceptance transfers waste custody for treatment. Product title follows the agreed sale. Toll processing retains household output ownership.

Table A2.1. Material, custody and payment handoffs.

Handoff	Responsible action	Payment or ownership
Commercial source → home	Customer separates. Route and home confirm accepted kg and destination.	Customer pays SIRKULA. Home earns handling payment.
Resident → home	Witness gross, tare, rejects and accepted kg.	SIRKULA funds the resident reward. No commercial fee.
Specialist → home	Identify starter age, lot, condition and receipt.	Starter, delivery and support have explicit payers.
Home → buyer	Buyer accepts grade, quantity and price.	Buyer pays home directly. SIRKULA invoices sales fee.
Home → processor	Contract specifies purchase or toll processing, recovery and rejects.	Raw sale transfers ownership. Toll service retains home ownership and charges processing.

Journeys cover eligibility, witnessed acceptance, production commissioning, service settlement and buyer acceptance. Table A2.1 identifies each handoff and payer.

Table A2.2. Initial treatment-service term sheet.

Term	Service obligation	Exception or record
Material and location	Identify eligible streams, contaminants and collection points.	Attach rejection examples. Cost remote stops.

Term	Service obligation	Exception or record
Minimum and allowance	Rp1,500,000 prepaid monthly. 200 accepted kg and daily collection.	Unused allowance expires. Reserved capacity and actual accepted kilograms are recorded separately. Plan billing requires an active paid service agreement.
Excess	Rp5,000/accepted excess kg. Settled one month later.	Confirm spare grow-out, finishing and route capacity. Bill accepted excess only under an active agreement. No excess invoice without a commercial customer.
Collection	Agree daily windows and same-day home delivery.	Record completed/missed visits and backup cost.
Weighing and remedy	Witness net accepted kg and corrections.	Rejects earn no handling. Remedy unperformed service.
Arrears and review	State due dates, escalation and notice before changes.	Keep unpaid balances. No hidden charges.
Service evidence	Source, accepted kg, date, batch and exception destination.	Monthly statement. Treatment records are not carbon certificates.

Reservations and accepted kilograms are separate. Handling follows commercial mass. Resident rewards generate no commercial invoice. Cost routes, support and timing together. Magalarva (n.d.-b) provides a service-process precedent.

Table A2.3. Household rights and approval controls.

Right or decision	Arrangement	Evidence/check
Voluntary entry and exit	Disclose workload, space, contribution, grant conditions, protection budgets and charges. Permit safe capacity reduction and orderly exit.	Reconcile equipment title, stock, balances and obligations. No wage or unconditional buyback guarantee. Normal business failure is not an automatic household loan.

Right or decision	Arrangement	Evidence/check
Household ownership	Signed equipment-grant and title handover identify the household owner and product seller.	Asset receipt, safe-use and misuse/exit terms, and accepted-lot/buyer records. No committed grant is inferred.
Charges and payment	Itemize earned/due/paid amounts. Finance prepares handling and rewards.	Second reviewer matches weights/payments. Consent before unrelated charges are netted.
Representation and appeal	Elected operators review rules. Assisted complaints receive independent review.	Decision dates and conflict declarations. No self-approval.
Prices and expansion	Governing body reviews costs with operator representatives. Fund assets separately.	Advance fee notice. Appendix 11.2 gates and technical/finance approval.

Disclose contribution, grant conditions, protection, charges and business risk. Reconcile assets, inventory and balances on voluntary exit. Normal trading failure is not automatic household debt. Operator representatives and independent review protect contested decisions without promising guaranteed income.

Restrict identities and finances by role. Retain changes and backups. Reconcile restored/offline records before payments or sales. Production needs authenticated permissions.

2.2 Linked records and platform prototype

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Table A2.4. Minimum transaction records and reconciliation.

Record	Required fields	Reconciliation
Identity and capacity	Date, source/home IDs, source type, contract, batch and route.	One load, one destination. Allocation fits capacity.
Intake	Gross, tare, rejects, accepted kg and corrections.	Accepted = gross − tare − rejects. Reweigh negative results.

Record	Required fields	Reconciliation
Resident reward	Accepted resident kg, rate, earned and paid dates.	Reward = accepted resident kg × rate.
Commercial service	Plan, allowance, schedule, accepted/excess kg, due and paid dates.	Invoice = plan + accepted excess × rate.
Home remuneration	Commercial kg, handling rate, earned and paid dates.	Handling = accepted commercial kg × rate.
Product sale	Lot, grade, moisture basis, accepted kg, price and buyer receipt.	Sales = accepted kg × price. Retain non-sale stock separately.
Charges and exceptions	Itemized fees, unpaid balances, complaint and correction IDs.	Reconcile earnings, receivables, cash and renewal saving.

Invoices remain receivables until paid. Household sales and SIRKULA fees remain separate. Corrections preserve the original entry, reason and reviewer. Each load and sold lot has one reconciled allocation.

The platform prototype demonstrates linked tasks and transaction views. Launch implementation adds authenticated access, capacity controls, ownership terms and correction history.

Table A2.5. Platform views and the work they support.

View	Participant task	Connected record
Resident	Check eligibility, offer material, confirm accepted weight and inspect reward.	Source, receipt, accepted/rejected mass and reward status.
Pangantara	Accept intake, allocate batches, record treatment and inspect product lots and statements.	Receipt, batch, harvest, stock and household settlement.
Commercial customer	Specify separated material, follow service evidence and inspect invoicing.	Agreement, treated batches, invoice and payment status.

View	Participant task	Connected record
SIRKULA team	Coordinate capacity, review quality, reconcile transactions and track exceptions.	Routes, approvals, held/released lots, corrections and balances.

Source: SIRKULA prototype interface and operating design.

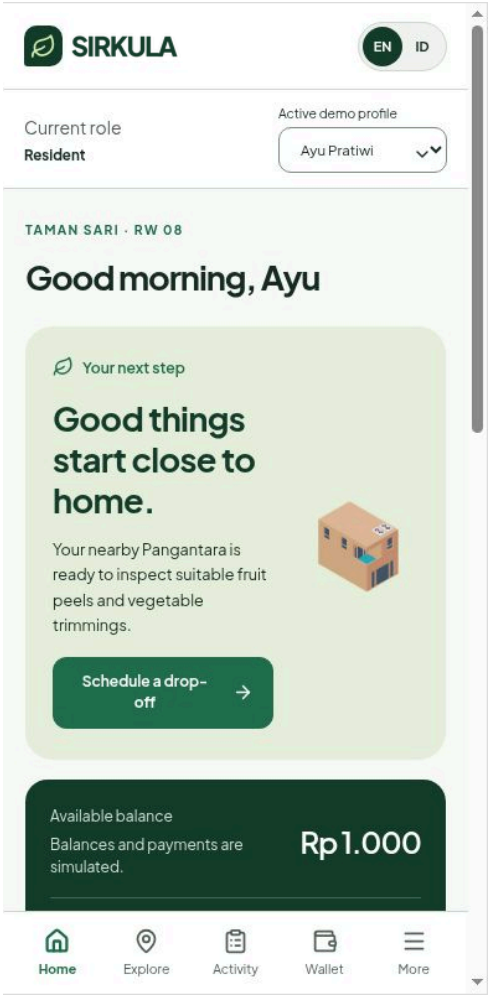


Figure A2.1. Resident drop-off and transaction navigation. Source: Actual SIRKULA platform interface with demonstration records.

The interface distinguishes intake, completed treatment, held or released stock, buyer acceptance and payment. Authenticated permissions, restricted financial views and recoverable records support operating use.

Appendix 3 Equipment workspace and specialist designs

3.1 Household installation equipment and storage

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Table A3.1. Four-rack installation envelope.

Function	Provision	Operating boundary
Covered work area	6 × 5 m clear interior = 30 m²	Existing covered work area. Wall thickness is outside the stated area. Washable floor and suitable services. Separate from living and food preparation.
Grow-out	Four GROW 10 racks, 40 trays	Each rack has a 1.40 × 0.80 × 1.65 m body envelope. Supplier checks confirm tray fit, safe access, loaded stability and usable capacity within that envelope.
Preparation and packing	Guarded powered chopper. Two 1.20 × 0.65 m tables	Dirty intake and clean dispatch are zoned. Sequential handling and cleaning manage the shared southern work route. Washwater is contained.
Residue finishing	Ten aerated 120 L bins	Five weekly cohorts, two bins each. Rain-protected aerobic finishing.
Raw and finished storage	Four 60 L raw bins. 150 kg finished-stock allowance	Date batches. Separate rejects and lindi. Dispatch before stock limits.
Access and handling	1.70 m rack-front route. 1.20 m curing aisle	Rack and curing aisles connect to a shared southern work zone. Stop cart movement during preparation or harvest. Keep door swings and exit paths free.

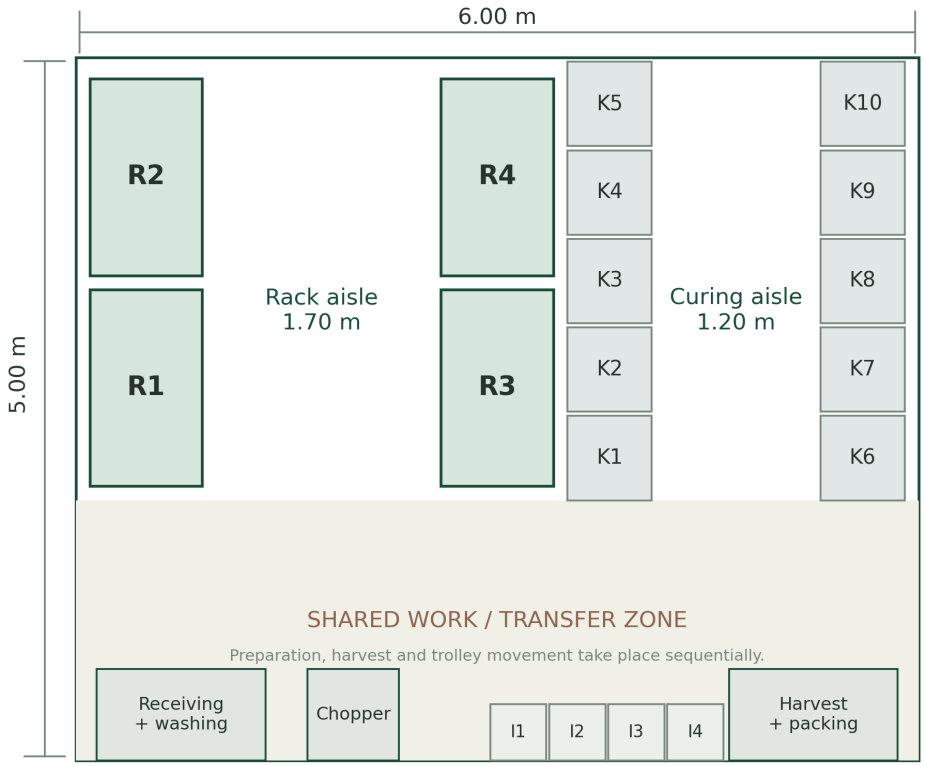


Figure A3.1. Dimensioned four-rack household workspace.

The existing covered interior is 6 × 5 m. Walls lie outside that area. Shared-zone preparation, harvesting and trolley movement are sequential. Keep exits clear and contain washwater and liquids.

Table A3.2. Four-rack equipment and minor-retrofit allowance.

Item group	Rp million
Four racks and 40 trays	10.40
Receiving and harvesting tables	1.50
Wash station and contained wet service	0.85
Ten mobile aerated curing bins	2.50
Raw bins rejects and captured-liquid vessels	0.65
Screens scales monitoring tools and trolley	1.50
Finished-stock cabinet	1.00

Item group	Rp million
Ventilation surfaces splash protection and PPE	2.20
Installation and contingency	1.40
Guarded powered chopper and stand	2.00
Total equipment and minor retrofit	24.00

2027 planning amounts. Rp million. Excludes land, a new building or slab, major service connections, cash rent, financing charges and working capital.

Rp2,000,000 preparation cost and 75 kg/hour effective chopping are planning assumptions. A published BSF-feed model is rated 50–100 kg/hour (CV Rumah Mesin, n.d.). Procurement requires a quote and representative-feed trial.

Confirm tray loading and fit, rack stability, reach, guarding, usable storage, lids, utilities and cleaning access. Tray dimensions and load ratings require coordinated supplier specifications.

Table A3.3. Storage and service fit at the two household scales.

Provision	Four-rack home	Twelve-rack home
Raw-feed storage	Four 60 L bins. 240 L total. Design 0.40 × 0.40 m footprints along the south wall.	Four wheeled 180 L bins. 720 L total. Design 0.50 × 0.60 m footprints along the east service wall.
Finished-stock storage	Closed cabinet under M2: at least 300 L usable capacity and 150 kg load rating.	Closed cabinet: at least 900 L usable capacity and 450 kg load rating. Design 1.20 × 0.65 m footprint.
Wet and clean stations	M1 integrates receiving and washing. M2 is the clean harvest/packing top over the stock cabinet. Shared-zone work is sequential.	Separate prep/wash and harvest/packing worktops. The 0.65 m standing strip is distinct from the 1.20 m movement route.

Provision	Four-rack home	Twelve-rack home
Handling and containment	Folded trolley under M1. Separate enclosed reject and captured-liquid vessels beside the chopper.	Park trolley and containment vessels in marked service bays. Empty large raw bins in small batches. Do not lift filled bins.

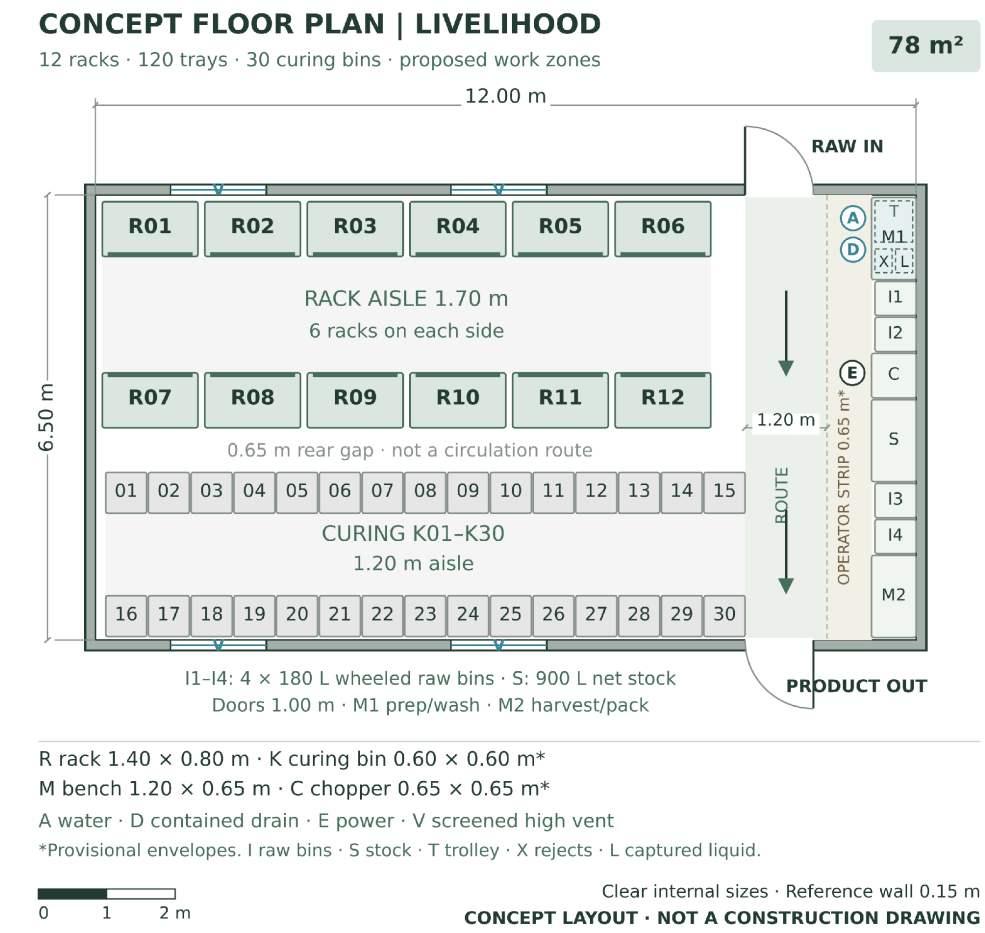


Figure A3.2. Separate twelve-rack enterprise workspace.

3.2 Grow out receiving and harvest equipment views

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The views retain the supplied equipment geometry and ten tray rack identity. Coordinated fabrication, load capacity and hygiene checks precede procurement.

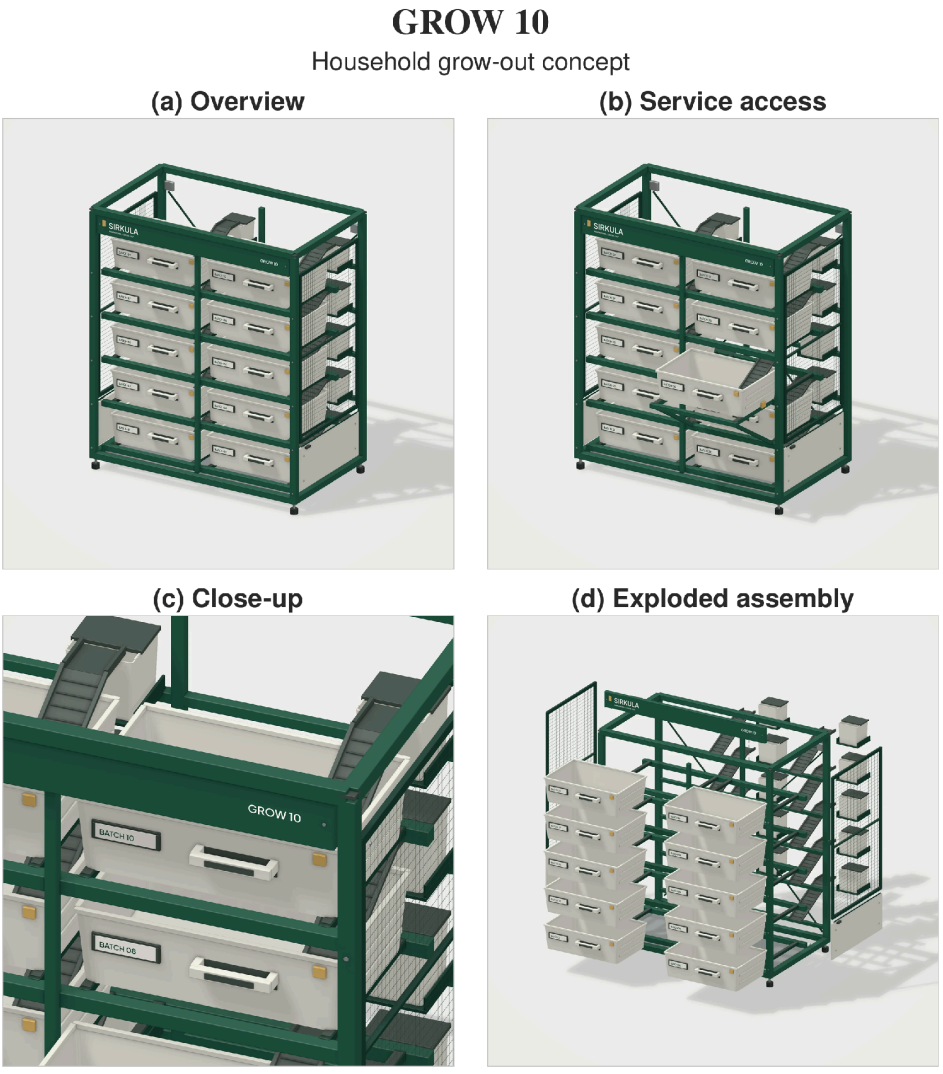


Figure A3.3. GROW 10 grow out rack views.

Household grow-out concept. Ten removable batch trays in two columns and five levels. Replaceable mesh panels, spill tray and adjustable feet. Removable ramps and rear collection cups.

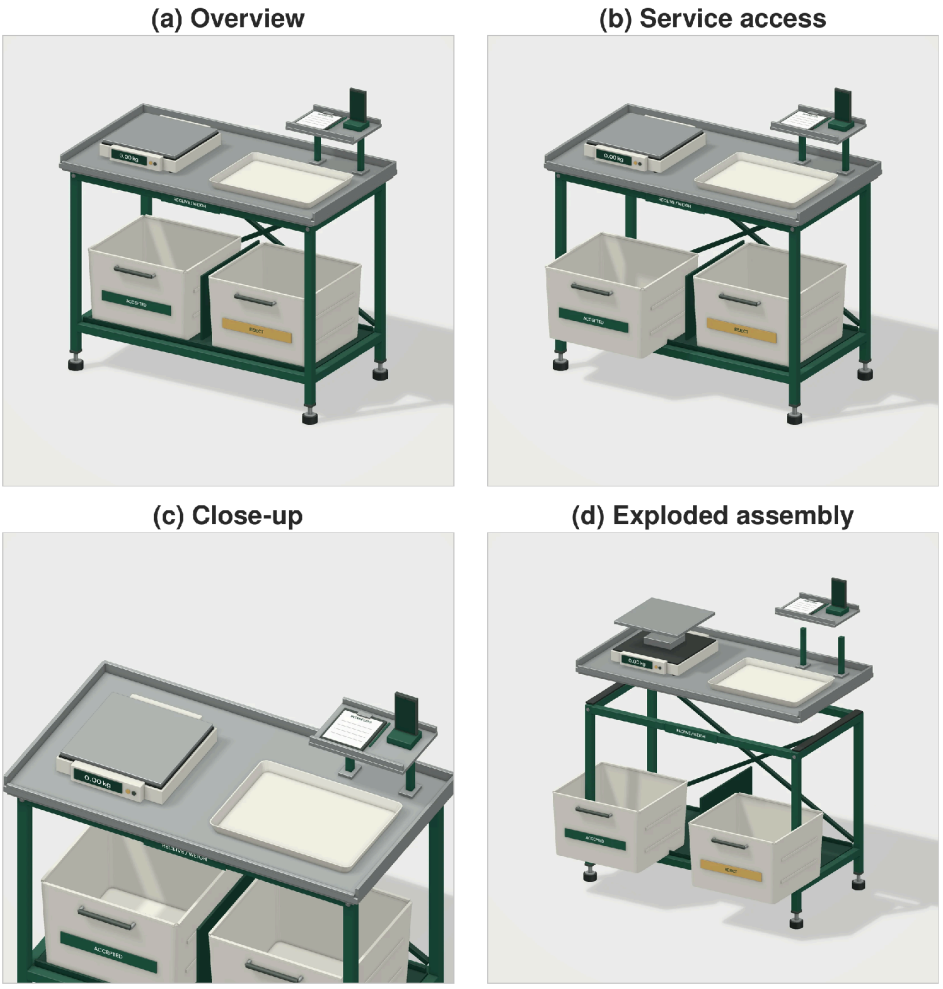


Figure A3.4. RECEIVE intake station views.

Household intake station concept. Platform scale with displays on both sides. Separate inspection vessel and worktop. Raised shelf for intake notes or a device.

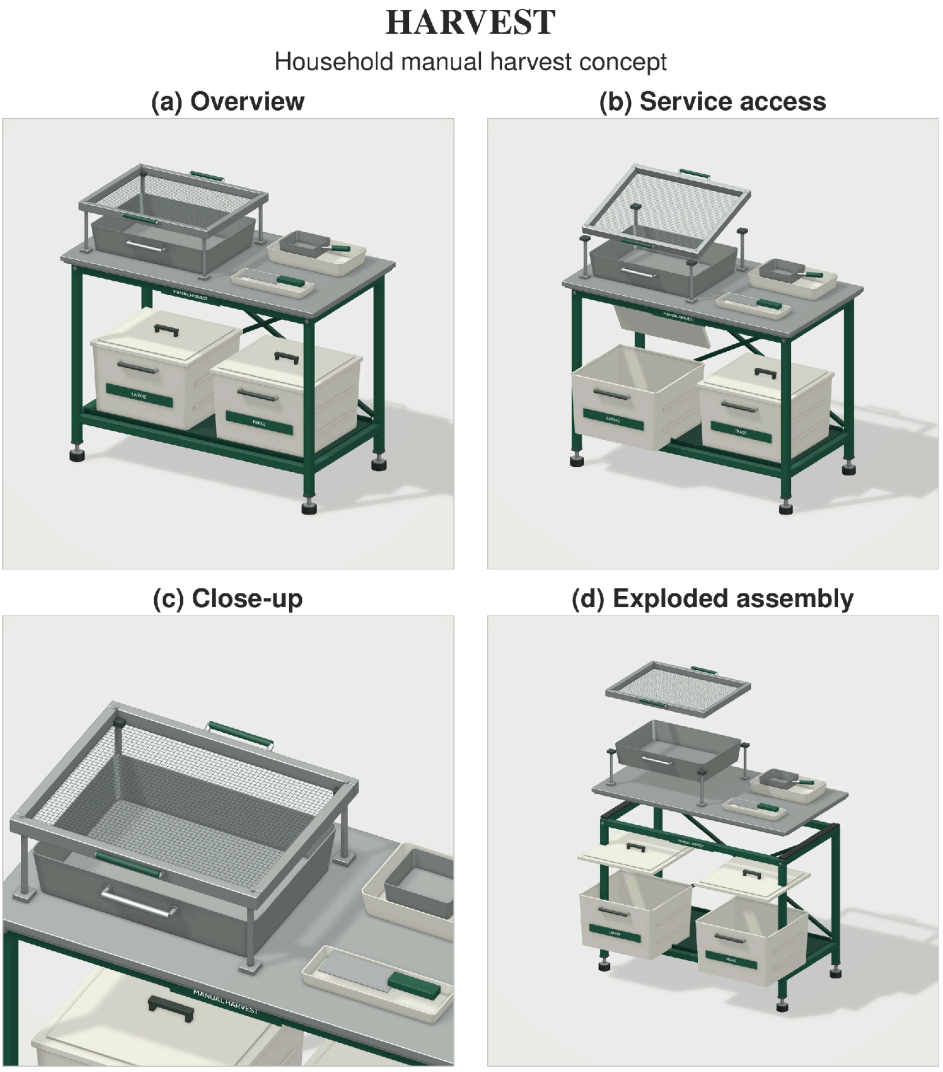


Figure A3.5. HARVEST manual station views.

Household manual harvest concept. Removable manual sieve above the worktop. Separate output vessels and basin. Accessible lower storage shelf.

3.3 Specialist breeding and nursery equipment views

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Breeding and nursery work is purchased from specialists at launch. These supplied designs support a later separately financed specialist facility.

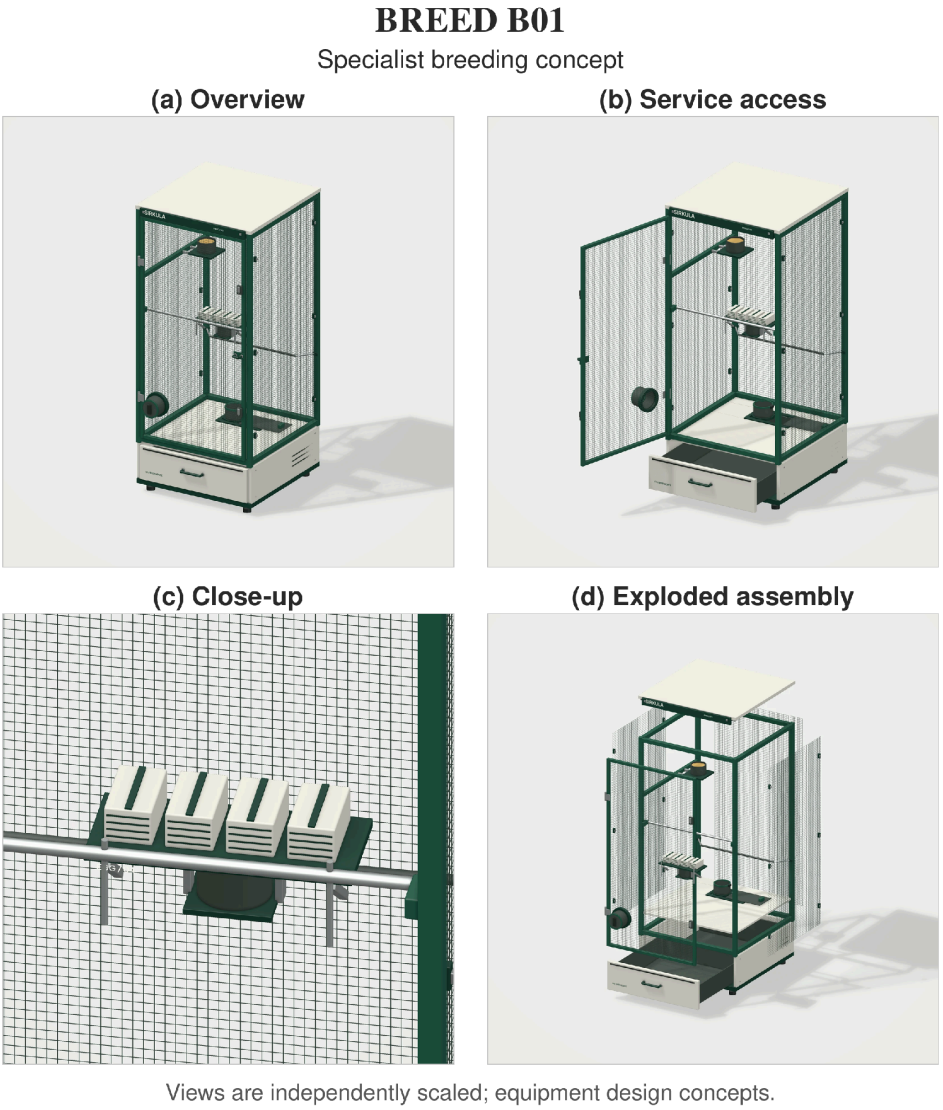


Figure A3.6. BREED B01 specialist breeding views.

Specialist breeding concept. Adult-flight mesh enclosure and shaded emergence area. Removable egg blocks on an accessible rack. Service sleeve, door and removable husbandry fixtures.

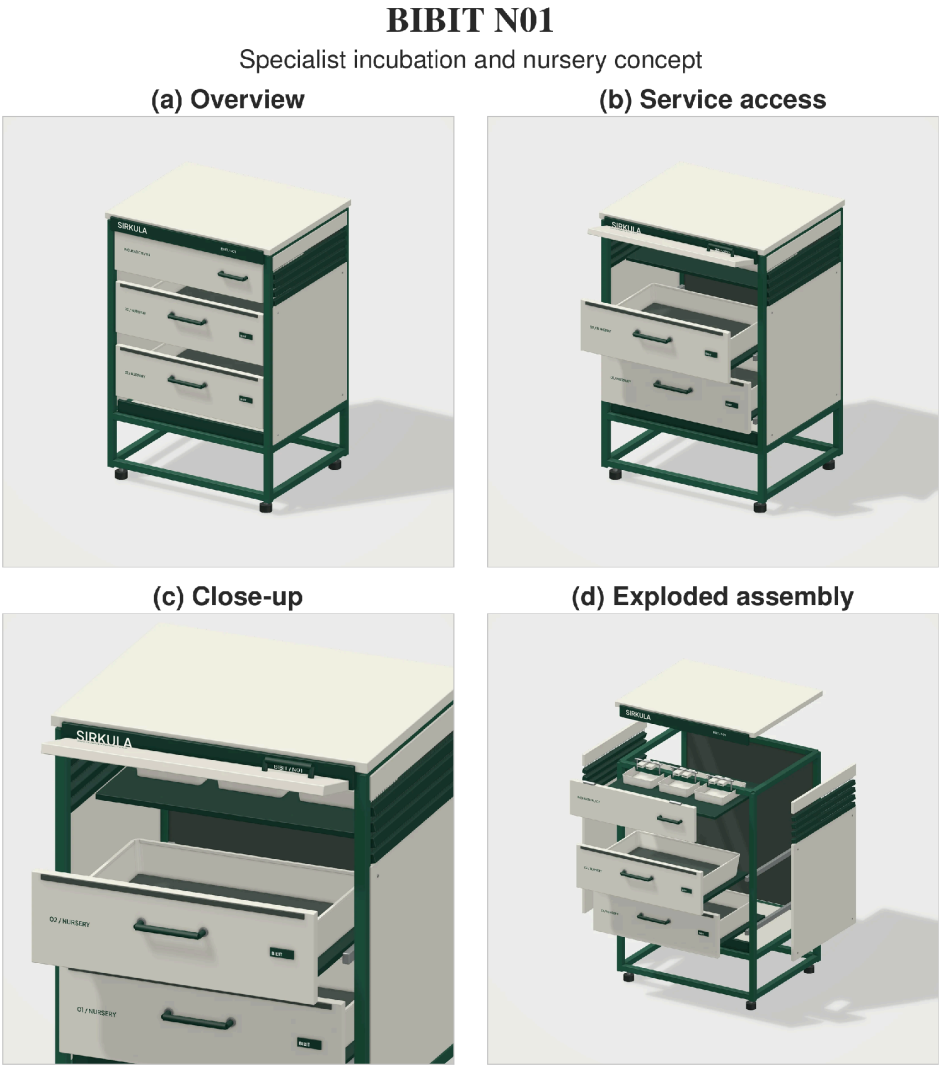


Figure A3.7. BIBIT N01 specialist nursery views.

Specialist incubation and nursery concept. Upper incubation access flap with removable egg supports and catch dishes. Two separate removable nursery drawers. Shaded panels and serviceable enclosure.

Appendix 4 Capacity material balance and operator work

4.1 Tray calendar and mature physical account

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Nominal capacity = 40 × 15 kg ÷ 18 days = 33.33 kg/day. Mature monthly feed = 33.33 × 85% × 365.25/12 = 862.40 kg. These are design assumptions. Dortmans et al. (2021) provide the process framework.

Table A4.1. Single-tray occupation schedule.

Day	Task	Feed and capacity rule
1–12	Inspect and feed daily	1.25 kg/tray/day. Cumulative 15 kg.
13–15	Observe late grow-out	No automatic additional ration.
16	Harvest and separate	Weigh all product and residue allocations.
17–18	Wash dry and inspect	Turnaround remains part of occupation.

Twenty tray pairs start over 18 days, with two pairs on days 1 and 10. Harvests involve two or four trays. Peak feed addition is 29.75 kg/day. Occupation of 15/20 days gives 40/30 kg/day. Target 75% moisture through measured blending. Added water earns no waste-treatment payment.

Table A4.2. Four-rack feed, product and managed-output account.

Stream	kg/month	Basis or destination
Accepted feed	862.40	80% commercial and 20% resident. Common tray capacity.
Gross fresh larvae	103.49	12% wet yield. 70% moisture assumption.
Retained larvae	2.07	2% of harvest. No sale receipt.
Fresh larvae sold	86.71	95% quality acceptance. 90% sell-through.
Wet residue	301.84	35% of feed. 55% moisture.
Finished kasgot	188.07	90% dry-matter recovery. 35% final moisture.

Stream	kg/month	Basis or destination
Finished kasgot sold	135.41	90% acceptance. 80% sell-through.
Captured raw lindi	17.25	2% of feed. Managed at cost, zero sales.

Wet kilograms unless a moisture basis is stated. All quantities use unrounded calculations. Displayed values are rounded.

Assumptions: 12% larval wet yield, 70% moisture, 2% harvest retention, 95% acceptance and 90% sell-through. Residue: 35% of feed at 55% moisture. 90% dry-matter recovery, 35% finished moisture, 90% acceptance and 80% sell-through. Raw liquid is 2% of feed.

Count biological/moisture losses, retention, rejects and unsold stock separately. Allocate each harvest once. Material assigned to processing or liquids is deducted from remaining product streams.

4.2 Complete work account protection and continuity

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Table A4.3. Mature monthly active work.

Task	Calculation	Hours
Routine checks records and machine cleaning	0.6333 h × average calendar days	19.28
Sorting preparation feeding and tray work	0.041667 h × accepted kg	35.93
Turning screening and hygiene handover	0.027 h × all finished kg	5.08
Packing and product dispatch	0.006 h × sold product kg	1.33
Total normal active work	All grow-out preparation and finishing	61.62
Planned high-load day	Staggered harvest and finishing	2.4

Daily routine is 30 minutes plus eight minutes for machine setup/wash. Variable work includes 0.2 minutes/kg sorting and 75 kg/hour chopping. Spread turning across five days. Screening on a two-tray harvest day avoids a 3.4-hour peak.

Table A4.4. Preparation-rate sensitivity.

Effective rate kg/hour	Total home h/month	Interpretation
50	67.37	Adds 5.75 hours/month to the base work budget.
75	61.62	Base design assumption.
100	58.75	Higher productivity case. Measure end-to-end work.

Table A4.5. Owner work, relief and binding limits.

Measure or case	Work basis	Required response
Normal mature work	61.62 owner hours/month. Two relief hours are standby and do not automatically reduce the normal task total.	Log owner and paid-relief hours separately. Cash income/hour uses actual owner time.
Economic work ceiling	Approximately 69.42 owner hours/month at current income and workspace valuation.	The nonnegative economic-residual gate can bind before the 75-hour supplementary-work ceiling.
25% longer task case	77.03 hours/month before relief. A complete two-hour substitution would leave about 75.03 owner hours.	Two funded relief hours alone do not demonstrate a passed 75-hour or economic-residual gate. Reschedule, reduce starts or approve additional costed help.
Declining delivered intake	Current floor: 753.46 total kg/month at base mix and prices, subject to updated local conditions.	Restore eligible paid volume or revise the route and starts. Never accept unsuitable material or claim reserved capacity as actual intake.

Source: Household task model and revised standby-relief policy.

Relief substitution requires recorded deployment and verified competency. Work and intake thresholds must be recalculated if income, source mix, task times or service terms change.

Table A4.6. Curing and finished-stock capacity.

Measure	Four-rack design
Available working volume	10 × 120 L × 80% = 960 L
Assumed wet bulk density	0.50 kg/L. Weigh a filled sample bin
Maximum seven-day cohort	71.40 kg. Two-bin capacity 96 kg
Maximum 35-day cycle stock	348.08 kg. About 696.15 L
Longer 42-day post-fill hold	Seven weekly cohorts. 14 bins plus space
Finished-product cabinet	150 kg model limit. Dispatch and age checks

Cohorts fill for seven days and hold 28 more. Capacity timing does not replace maturity, hygiene and buyer release. Full stock or delayed qualification reduces intake. Record source age. Deliver and prepare daily.

Appendix 5 Products service markets and tariffs

5.1 Products processing and direct household sales

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Fresh larvae uses the December 2024 Rp4,000/kg reference (Tohari, 2024); finished kasgot assumes Rp1,500/kg. Magalarva's product and buying routes identify prospective outlets for product qualification (Magalarva, n.d.-a, n.d.-c, n.d.-d).

Table A5.1. Product routes and the commercial work they require.

Product route	Processing and market requirement	Launch decision
Fresh larvae	Accepted and traceable whole larvae. Agreed handling, quantity, delivery and grade.	Base household sale. Rp4,000/kg reference with acceptance and sell-through losses.
Finished kasgot and media blends	Controlled finishing, screening, hygiene, moisture and crop-use specification. Blends add purchased ingredients.	Finished kasgot base sale at assumed Rp1,500/kg. Blends are a separately costed alternative.
Eggs, starters and breeding	Adult colony, egg handling, nursery survival, dedicated space and scheduled supply.	Purchased specialist inputs at launch. Later specialization requires its own cost and demand case.
Dried larvae, meal and formulated feed	Fresh allocation, drying/conversion losses, processing or toll charges, specification and qualified buyers.	Contract drying comparison retained. Downstream processing revenue belongs to the actor selling that product.
Oil, fats and oleochemicals	Extraction/refining and specialist product testing. Meal and oil may share one allocated batch.	Future processor partnership. No base household receipt.
Liquids, energy and solid-residue alternatives	Defined formulation or qualified energy/thermal outlet. Allocate feedstock and cost treatment, testing and rejects.	Raw lindi managed at cost. Liquid, biogas and biochar routes require their own accepted use.

Product route	Processing and market requirement	Launch decision
Chitin, melanin, peptides and human-food research	Specialist extraction or a separate food-grade chain, purity, safety, approvals and accepted buyers.	Research and specialist development. Excluded from base household cash.

Offers specify grade, moisture, quantity, price, delivery and rejects. Raw sale transfers title on acceptance. Toll processing retains household title. Processor revenue is distinct from farm-gate receipts.

Crop and hygiene findings are treatment-specific (Fauzi et al., 2022. Suwirmen et al., 2022. Tan et al., 2021. Van Looveren et al., 2022). Liquids need a defined composition and use. Retain both liquid and solid streams.

Specialist pathways establish prior art, not household receipts (Protix, n.d.-a, n.d.-b. Innovafeed, n.d.. Insectta, n.d.. Riolo et al., 2023. Siqueira Galvão Novo et al., 2024).

Table A5.2. Sale and processing choices.

Arrangement	Household receipt	Costs and ownership
Fresh larvae	Accepted kg × raw price.	Home pays production/delivery. Title transfers on agreed acceptance.
Toll drying	Accepted dried kg × dried price.	Home retains title. Cost raw allocation, losses, toll, packing and delivery.
Kasgot and blends	Accepted finished kg × product price.	Cost maturation, screening, rejects and storage. Blends add materials and mixing.
Liquid or specialist route	Contracted raw or qualified finished-product receipt.	Define title and processing liability. No automatic downstream income.

Test orders establish prices and volumes. Record household buyer receipts and the separate 10% SIRKULA sales fee with earned, due and paid dates. Reconcile unpaid balances on exit.

5.2 Treatment service market size

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Table A5.3. Treatment service TAM SAM and SOM.

Layer	Geography and scope	Argumentation
TAM	Eligible food source establishments across Depok	The 2024 registry contains 1,713 establishments. An illustrative 10–30% eligibility screen gives 171–514 accounts and Rp3.078–9.252 billion annual base fees at one 200 kg plan per account.
SAM	One proposed Sukmajaya route cluster	Screen 30–50 addresses and shortlist 10–20 eligible prospects. Ten to twenty paid annual plans would represent Rp180–360 million in base fees.
SOM pilot	One commissioning home then one mature home	Capacity permits one plan at commissioning and three 200 kg plans at maturity. Steady annual base fees are Rp18–54 million, depending on active paid agreements.
SOM endpoint	Nineteen mature homes in the ten year model	A maximum of 57 reservations would give Rp1.026 billion annual base fees in constant 2027 prices at full year maturity. Forecast cash follows actual commissioning and escalation.

Market definitions follow Faris et al. (2024). The establishment registry uses 2024 data (Dinas Kesehatan Kota Depok, 2025). Eligibility shares, shortlist sizes, tariff and capacity are author planning estimates; annual amounts are gross base fee illustrations, separate from excess fees, costs and product receipts.

5.3 Product outlet market size

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Table A5.4. Product outlet TAM SAM and SOM.

Layer	Geography and scope	Argumentation
TAM	Regional aquaculture and feed buyers for larvae and growers or media retailers for kasgot	Size each product market from compatible annual purchase volumes and accepted prices. Fish output and crop area help locate prospects but do not measure BSF product purchases.
SAM	Buyers reachable within product handling and quality requirements	Interview 6–10 prospects per product route and sum volumes meeting grade, timing, transport and payment requirements. Magalarva provides prospective outlet precedents.
SOM one home	One mature four rack household	Modeled sales are up to 86.71 kg fresh larvae and 135.41 kg finished kasgot per month, limited by accepted paid orders. Fresh larvae uses the Rp4,000/kg December 2024 reference; kasgot assumes Rp1,500/kg.
SOM nineteen homes	Nineteen mature four rack households	Supply reaches up to 1,647.53 kg fresh larvae and 2,572.78 kg finished kasgot per month. This is a mature supply ceiling; phase in follows the monthly model.

Sukmajaya produced 118.57 tonnes of freshwater grow out fish in 2025, within 1,962.91 tonnes citywide (Badan Pusat Statistik Kota Depok, 2026, Table 5.3.2). These sector figures guide prospect discovery. Product purchase volumes and prices are established through buyer qualification; modeled supply and market demand remain separate.

5.4 Service tariffs and market implementation

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Table A5.5. Public tariff and service-scope comparison.

Service	Price	Essential scope
Depok restaurant or café municipal service	Rp50,000/m³	Municipal restaurant/café category. Tariff row does not specify daily organic collection or frequency.
Depok commercial container transport	Rp750,000/month	Commercial premises. Listed row does not state container capacity or collection frequency.
W4C residential collection	From Rp30,000 per home per month excluding VAT	Domestic waste; 1–2 collections weekly, at least 100 homes and a one year term. Depok is covered.
W4C business RWTL	Quotation	Separated business organic and other domestic streams; package/monthly volume and at least 12 months. Request a matching daily service quote.
W4C Personal Waste Management	Not an organic comparator	Weekly inorganic collection in Jakarta/Tangerang. Organic material is excluded.
SIRKULA proposed commercial plan	Rp1,500,000/month for 200 kg plus Rp5,000 per accepted excess kg	Daily collection and accepted organic treatment in the proposed Sukmajaya cluster. An active paid agreement governs reserved capacity and excess.

Tariff sources: Pemerintah Kota Depok (2025), Appendix II, printed pp. 19–20; Waste4Change (n.d.-a, n.d.-b, n.d.-c), accessed 12 October 2026. Units, frequency and customer scale differ. Without measured density or an explicit mass allowance, these prices cannot be converted into a comparable per kg rate. W4C denotes Waste4Change.

Table A5.6. Market implementation records.

Activity	Record	Decision
Map eligible kitchens and route fit	Audit 30–50 addresses: source activity, location, service, age and separation.	Build a deduplicated eligible shortlist.
Measure material and variation	Weigh accepted and rejected organics for seven consecutive days; extend to 14 days where practical.	Agree delivered mass and collection windows.
Price daily service	Obtain three scope matched written business quotes and candidate current bills. Record a paid trial and renewal.	Set scope and price using payments and renewal.
Qualify product outlets	Obtain specification, moisture basis, test lot, payment and repeat orders for each product.	Schedule output against qualified paid demand.
Time full operator work	Record complete cycles, yield, starter/source age, finishing, sales work and incidents.	Keep intake within work and safety limits or fund qualified support.

The proposed audit, paid service trial and qualified product lots establish the inputs for commissioning and subsequent growth.

Appendix 6 Participant value and strategic priorities

6.1 Participant value

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Table A6.1. Value proposition by participant.

Participant	Need	SIRKULA value
Pangantara	Earn income from a suitable household activity with manageable work.	Owned equipment and output, paid handling, buyer coordination and inspectable settlement.
Commercial source	Remove separated organics reliably and account for treatment.	Scheduled collection, measured acceptance, a monthly statement and service escalation.
Resident contributor	Know what to separate and what was accepted.	Eligibility guidance, witnessed weighing and a funded reward.
Product buyer	Obtain consistent material with a known origin.	Lot specifications, accepted quantities and a direct household sale.

Compare complete service scope and bills with local collectors, composters and BSF processors. Paid trial and renewal establish customer value.

6.2 Strategy and planned capabilities

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Table A6.2. PESTEL implementation priorities.

Factor	Weight rating and score	Planned response
Political Municipal service context and coordination	Weight 0.08 Priority 4 Score 0.32	Confirm DLHK/kelurahan service interface and keep launch permission-dependent.
Economic Daily service scope and competing tariffs	Weight 0.26 Priority 5 Score 1.30	Obtain matched daily quotes, customer cost records, paid trial and renewal.

Factor	Weight rating and score	Planned response
Social Operator workload and neighborhood acceptance	Weight 0.16 Priority 4 Score 0.64	Time complete tasks, consult neighbors, document complaints and safe exit.
Technological Rack yield owner work and starter reliability	Weight 0.24 Priority 5 Score 1.20	Run complete cycles with starter traceability, source age and task diaries.
Environmental Contamination, odor, water and residue controls	Weight 0.16 Priority 4 Score 0.64	Reject unsuitable feed, contain liquids and reduce intake when storage binds.
Legal Current tariff, site, contract and product checks	Weight 0.10 Priority 4 Score 0.40	Verify applicable permits and buyer release standards before accepting or selling.
TOTAL	Weight 1.00 Priority — Score 4.50	Price and process account for 0.50 of total weight.

PESTEL categories follow Chartered Institute of Personnel and Development (2025). Author weights sum to 1.00. Priority 1 is minimal, 2 low, 3 routine, 4 must manage at pilot and 5 an immediate validation blocker. Weight × priority totals 4.50/5, indicating implementation attention rather than attractiveness or probability. Priorities concentrate on daily service pricing, complete cycle performance, funded capacity and neighbourhood conditions.

Table A6.3. Internal capabilities and implementation priorities.

Factor	Weight rating and score	Planned response
Physical/cash accounts and capacity constraints S	Weight 0.10 Response 3 Score 0.30	Retain reconciled material and cash logs.
Linked intake, lot and settlement prototype S	Weight 0.08 Response 2 Score 0.16	Implement authenticated roles and correction history.
Household title and participation controls S	Weight 0.08 Response 3 Score 0.24	Document title, charges and orderly exit.
Modular four-rack/outsource launch design S	Weight 0.08 Response 3 Score 0.24	Keep specialist assets outside household startup case.
Site, guarding and procurement checklist S	Weight 0.06 Response 2 Score 0.12	Obtain coordinated specs and timed representative-feed trials.
Paid daily service and renewals W	Weight 0.20 Response 1 Score 0.20	Require paid contract plus delivered-mass and renewal records.
Complete cycle and task performance W	Weight 0.14 Response 1 Score 0.14	Validate yield, storage, hours and safety through complete cycles.
Equipment and support financing W	Weight 0.12 Response 1 Score 0.12	Name funder, instrument, loss allocation and release conditions.

Factor	Weight rating and score	Planned response
Operating permissions and recovery W	Weight 0.08 Response 2 Score 0.16	Test permissions, backups and restored-record reconciliation.
Product orders and site quality W	Weight 0.06 Response 2 Score 0.12	Secure paid test lots and buyer acceptance before sales expansion.
TOTAL	Weight 1.00 Response — Score 1.80	Close paid-demand, performance and funding gaps first.

S identifies a design strength and W an implementation priority. The David et al. (2020) response scale is 1 poor, 2 average, 3 above average and 4 superior. Importance weights sum to 1.00; weight × response sums to 1.80, below the heuristic 2.50 midpoint. These author ratings organise implementation work rather than report field or stakeholder scores.

Table A6.4. External factors and planned responses.

Factor	Weight rating and score	Planned response
Urban concentration may support compact route O	Weight 0.08 Response 2 Score 0.16	Map actual clustered source addresses (Badan Pusat Statistik Kota Depok, 2026).
Registered food-premises prospect frame O	Weight 0.10 Response 2 Score 0.20	Screen registry by location, material and mass (Dinas Kesehatan Kota Depok, 2025).
Organic collection services already offered in Depok O	Weight 0.08 Response 2 Score 0.16	Request comparable business quote (Waste4Change, n.d.-b).

Factor	Weight rating and score	Planned response
Fresh/frass product and buyer pathways exist O	Weight 0.10 Response 2 Score 0.20	Test grade, lot size and payment (Magalarva, n.d.-a, n.d.-c, n.d.-d).
Local BSF training provides implementation precedent O	Weight 0.08 Response 2 Score 0.16	Contact qualified specialists without assuming capacity (Tohari, 2024).
Lower-priced public and residential alternatives T	Weight 0.18 Response 1 Score 0.18	Demonstrate paid value for daily commercial scope (Pemerintah Kota Depok, 2025).
Source age, contamination and inconsistent delivery T	Weight 0.12 Response 2 Score 0.24	Use intake eligibility, daily preparation and backup routes (Dortmans et al., 2021).
Hygiene and buyer qualification limit saleable output T	Weight 0.10 Response 2 Score 0.20	Hold/reject lots until evidence supports release (Dortmans et al., 2021).
Tariff and applicable-rule changes T	Weight 0.08 Response 2 Score 0.16	Check current legal/fee version and document supplier scope (Pemerintah Kota Depok, 2025).
Odor, nuisance and household acceptance constraints T	Weight 0.08 Response 2 Score 0.16	Inspect containment and respond through intake reduction (Dortmans et al., 2021).

Factor	Weight rating and score	Planned response
TOTAL	Weight 1.00 Response — Score 1.82	Verify price acceptance, mass and qualified outlets.

O identifies an opportunity and T a threat; ratings describe the planned response, rather than a factor's desirability. The same 1–4 method gives total weight 1.00 and score 1.82. Method follows David et al. (2020) and David (2025); related factors are not statistically independent.

Appendix 7 Registration commissioning and partnerships

7.1 Registration and commissioning pathway

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Table A7.1. Proposed Pangantara registration and acceptance checklist.

Stage	Applicant or team evidence	Decision and responsible function
Application and consent	Applicant identity, contact, voluntary participation and permission to use an existing household workspace. Assisted registration is available.	Operations records the application. No equipment purchase or household debt is required at application.
Workspace screening	30 m² covered clear interior, suitable access and services, separation from living/food preparation, contained washing and agreed neighbourhood conditions.	Technical lead records site fit and required corrections. Unsuitable sites receive a reasoned decision.
Time and task readiness	Declared owner availability, practical handling capability, training participation and the intended supplementary-work limit.	Operator and technical lead agree the full task plan and relief escalation. No guaranteed income is offered.
Capital and timing	Rp2,400,000 operator contribution and Rp361,146 working cash. Proposed Rp21,600,000 grant allocation is funded before procurement.	Finance verifies funding and disclosures. Application or acceptance is not evidence of a secured sponsor.
Demand and supply	Seven-day source audit, paid service offer, acceptable product outlets and priced starter/technical/route support.	Operations and technical lead confirm capacity and demand fit for that home.
Agreement and protection	Signed equipment-grant/title, safe-use, fees, product ownership, voluntary exit, incident and social-protection arrangements.	Finance and operator complete the terms before commitment. Ordinary failure does not automatically create a loan.

Stage	Applicant or team evidence	Decision and responsible function
Training and setup	Intake, guarding, preparation, batch allocation, abnormal lots, harvest, finishing, cleaning and a reconciled statement.	Technical lead commissions the home only after demonstrated readiness.
Operating review	First four paid weeks, initial three-month production record and reviews at months 6 and 12.	Operators remain, correct scope, suspend safely or exit according to the funded household and service gates.

Assisted registration and consistent screening support the household acceptance pathway. Operations records decisions and appeals, while site, funding and practical competencies are completed before commissioning.

Table A7.2. Work packages, accountable functions and completion evidence.

Work package	Lead and output	Release condition
Market and route	Commercial coordinator: seven-day source audit and paid offer.	Renewal and delivered kg support the reserved route.
Product outlets	Home seller and sales coordinator: grade, price and test orders.	Repeat orders support 86.71 fresh kg and 135.41 kasgot kg/month.
Home and equipment	Technical lead: measured site, specifications, quotes and title.	Complete kit fits an existing suitable 30 m² area. Funding agreed.
Specialist service	Technical lead: starter calendar, QA, backup and remedies.	All visits, testing and emergency response costed.
Training and first cycle	Operator and trainer: witnessed handling and timed work.	Safe preparation, full cycle, cleaning and records meet the work budget.
Settlement and review	Finance with operator representative: receipts, income/hours and service record.	Clear unexplained balances. Apply Appendix 11.2 gates.

Commissioning follows source measurement, route/buyer terms, home fitness and funding. Review paid service after four weeks, production after three months, and continuing outcomes at months 6 and 12.

7.2 Work schedule and operator competencies

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Table A7.3. Proposed implementation schedule and completion evidence.

Work package	Relative timing	Dependency and evidence
Source audit and assisted applications	Preparation week 1 to Preparation week 2	Approved proposed catchment and access to prospective sources. At least seven consecutive days of eligible material, rejects and current collection practice
Service and buyer discussions	Preparation week 1 to Preparation week 3	Measured source streams and prospective buyer specifications. Priced trial offer, buyer grade/quantity/payment terms and route quotation
Site, grant and funding approval	Preparation week 2 to Preparation week 4	Applicant/site assessment and external funding decision. Qualified site, signed title/grant terms and actual funded allocation
Procurement, installation and training	Preparation week 4 to Preparation week 6	Site and funded capital approval. Complete equipment fit, safe handling competencies and starter/support calendar
First paid service and settlement test	Operating month 1 to Operating month 1	Commissioned ready household, paid service agreement. First four paid weeks of service, weights, work, payments, complaints and renewal
Initial production and product qualification	Operating month 1 to Operating month 3	Qualified household and continuous starter/route support. At least three completed grow-out cycles, two paid accepted larvae lots and one qualified kasgot sale

Work package	Relative timing	Dependency and evidence
Funded-cohort review and conditional additions	Operating month 3 gate to Operating month 12 review	Household, paid-demand, quality, protection and funding gates. Month 3 first review, month 6 and 12 follow-ups. Additions only after a passed and financed review

Preparation weeks are planning estimates before paid month 1. Commissioning tests are not guaranteed biological outcomes. Tasks wait for their approval and funding prerequisites.

Table A7.4. Training and witnessed competency record.

Practice	Observable evidence	Response to failure
Terms and settlement	Explain asset/output title and all statement charges.	Clarify before financial commitment.
Intake and preparation	Demonstrate weighing, guarding, cleaning and feed allocation.	Correct unsafe practice or reduce intake.
Husbandry and abnormal lots	Identify moisture, odor and starter problems. Isolate a lot.	Suspend additions. Call technical support.
Harvest and finishing	Weigh allocations. Distinguish raw residue from finished product.	Hold nonconforming lots.
Hygiene and records	Separate tools/water from household food. Preserve corrections.	Correct layout or records before release.

Demonstrate a complete cycle, timed work and reconciled settlement. Correct unsafe practice and hold nonconforming products before commissioning.

7.3 Partnership terms responsibility and approval

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Table A7.5. Partnership terms and responsible functions.

Function	Terms to establish	Accountable actor
Breeder and technical adviser	Starter age, quantity, delivery, training, QA and backup.	Technical lead.

Function	Terms to establish	Accountable actor
Route contractor	Daily stops, windows, weighing, home delivery and diversion.	Operations coordinator.
Processor or crop-input specialist	Title, recovery, hygiene, testing, toll price and rejects.	Product/technical coordinator.
Product buyer	Grade, quantities, price, delivery and payment.	Home seller with sales coordinator.
Community and finance providers	Community communication, conditional equipment-grant funding, signed title and safe exit terms.	Operations and finance. No sponsor is committed.

Launch support is outsourced. Qualified people may combine functions. All task hours and services remain costed. An owned hub needs separate capacity, occupancy, staffing, maintenance and capital analysis (Appendix 8).

Table A7.6. Responsibility and approval functions.

Activity	Accountable approval	Responsible functions
Application and screening	Operations	Operations and technical team; applicant consulted.
Site and technical readiness	Technical lead	Household and contracted support; operations/finance consulted.
External capital award	Funding partner	Finance; award precedes allocation.
Equipment grant and title	Finance	Operations and finance; household and reviewer consulted.
Service contract and route	Operations	Operations; household finance and support consulted.
Installation and training	Technical lead	Household and contracted support; operations consulted.
Household intake	Pangantara	Pangantara; operations and technical lead consulted.

Activity	Accountable approval	Responsible functions
Starter and specialist delivery	Technical lead	Contracted support; operations and household consulted.
Product release	Technical lead	Household and support; operations consulted.
Household product sales	Pangantara	Household and operations; finance/technical lead consulted.
Monthly settlements	Finance	Operations and operator reviewer; household consulted.
Qualified relief within budget	Operations	Operations and contracted support; household technical lead and finance consulted.
Contested charge or incident	Independent operator reviewer	Operations technical lead and finance; household/support consulted.
New funded cohort	Operations	Technical lead and finance; household reviewer and support consulted.

Accountability belongs to one approval function for each activity. Qualified people may combine paid functions, while contested charges retain independent review. Funding partners approve actual awards; paid route and specialist work, relief and incident review retain their stated budgets.

Appendix 8 Household economics coordinator costs and funding

8.1 Household account capital and distinct enterprise options

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Table A8.1. Four-rack household receipts and available income.

Receipt or cost	Rp per month
Fresh larvae sold	Rp346,849
Finished kasgot sold	Rp203,114
Commercial handling remuneration	Rp2,552,692
Total receipts	Rp3,102,654
Sales-service fee	−Rp54,996
Direct cash costs	−Rp645,393
Cash margin before renewal saving	Rp2,402,265
Equipment replacement saving	−Rp291,745
Available owner income	Rp2,110,520
Active work h/month	61.62
Income per active hour	Rp34,250
Residual after owner time and space	Rp194,999

2027 rupiah, average month = 365.25/12 days. Income before personal tax and financing charges. Economic appraisal values owner time at Rp25,000/hour and existing workspace at Rp375,000/month. Calculations use unrounded values.

Table A8.2. Direct household cash-cost drivers.

Cost	Calculation
Starter supply	10,000 per nominal 15 kg. Rp300/1,000. Rp172,479/month.
Utilities hygiene and machine upkeep	Rp200,000/month, including a Rp50,000 PPE/hygiene planning allocation and Rp150,000 utilities/upkeep. Existing workspace has no cash rent. Confirm actual prices before intake.

Cost	Calculation
Consumables	Rp150 × accepted feed kg.
Fresh packing and delivery	Rp300 × sold fresh kg.
Kasgot finishing and hygiene	Rp400 × all finished kg, including unsold output.
Kasgot packing and delivery	Rp250 × sold finished kg.
Rejects and raw liquids	Rp100 × managed non-sale and liquid kg.

Replacement saving remains earmarked household cash. Replacement uses that balance without a second income deduction. Owner time and space remain separate economic costs. SIRKULA pays the new social-protection and relief allowance, so household available income is unchanged. Distinguish earned income, receipts and balances.

Table A8.3. Conditional equipment grant and participant risk.

Issue	Proposed arrangement	Record and financing boundary
Equipment capital	Rp24 million package. Operator contributes Rp2.4 million and proposed conditional grant provides Rp21.6 million.	Grant funding is not secured. Committed-grant receipt inputs remain zero. Finance approves a real allocation before procurement. The kit grant is distinct from total startup and reserve funding.
Grant instrument and title	Conditional equipment grant rather than household debt. Equipment ownership is documented with household handover.	Safe use, record obligations, misuse remedies and exit terms are disclosed in the signed agreement. Ordinary business failure does not automatically become loan repayment.
Operator exposure	Own contribution, working cash, variable costs and income depend on actual production and demand.	No salary, unconditional product buyback or guaranteed earnings are promised. Keep household reserves and receivables separate.

Issue	Proposed arrangement	Record and financing boundary
Funding-pool exposure	The funder bears nonrecoverable grant capital. SIRKULA bears agreed coordinator/service obligations. Operator contribution and income remain at risk.	Prospective CSR/TJSL or philanthropic providers are uncommitted. Financing at SIRKULA level needs its own agreement and cash-flow treatment.
Failure and voluntary exit	Suspend safely, reconcile stock, equipment title, unpaid amounts and earmarked balances, and settle defined obligations.	Apply agreed misuse/exit rules without converting lawful trading failure into undeclared household debt. Record any actual recovery separately.
Capital and recurring protection	Grant capital is a funding allocation. JKK/JKM and standby relief are recurring SIRKULA costs.	Do not treat either as extra product or treatment revenue. NPV uses unfinanced cash. Zero terminal recovery remains the model basis unless changed explicitly.

This is the proposed operating term sheet, not a signed grant or legal award. Counterparty and final agreement terms are prerequisites to commissioning.

Operator equipment contribution is Rp2.4 million plus Rp361,146 working cash. Proposed conditional grant support is Rp21.6 million and is uncommitted. Funding precedes procurement. Contribution and cash-envelope recovery are model results, not a funding award or guaranteed reimbursement.

The twelve-rack case adds Rp38.5 million equipment, 48 m² and approximately 84.69 owner hours/month. Its mature household comparison is separate from the four-rack forecast and historical trial coverage. The Rp5.75 million contextual comparator rounds Rp5,522,662 × 1.04 = Rp5,743,568.48 (Gubernur Jawa Barat, 2025). This is an assumed 2027 comparison, not an official 2027 wage or enterprise guarantee.

Table A8.4. Two distinct household enterprise options.

Mature comparison at 2027 prices. The twelve rack option is separately costed; only four rack homes enter the base monthly rollout.

Measure	Four racks	Twelve racks
Racks / trays / workspace	4 / 40 / 30 m²	12 / 120 / 78 m²
Equipment package	Rp24,000,000	Rp62,500,000
Available income/month	Rp2,110,520	Rp6,547,043
Active work/month	61.62 hours	146.31 hours
Income/active hour	Rp34,250	Rp44,748

Source: Authors' equipment, workload and mature-household cost model. The twelve-rack case is a separate static illustration.

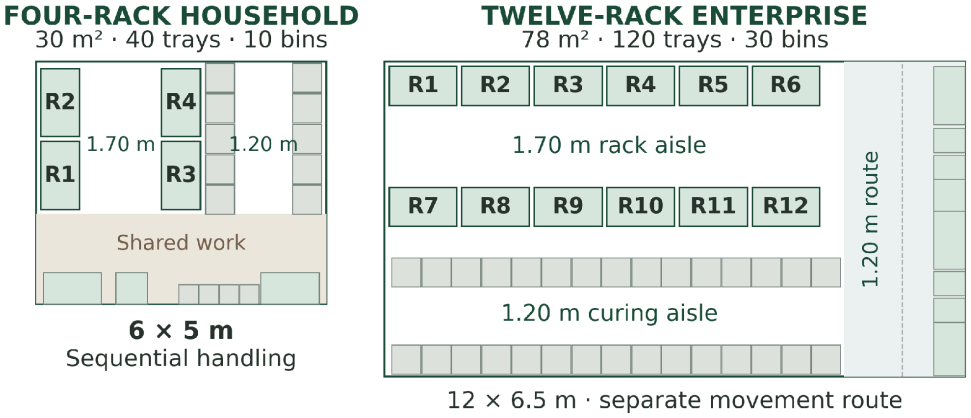


Figure A8.1. Workspace requirements at the two household enterprise scales.

Table A8.5. Optional processing mix and the starting fresh-product route.

Measure	Fresh and kasgot	Contract-processed mix
Income Rp/month	Rp2,110,520	Rp2,063,409
Work h/month	61.6	68.4
Kit Rp million	24.00	25.20

Separate mature comparison at 2027 prices. Optional mix is excluded from the four-rack rollout.

The mix allocates 20% of larvae to drying and 30% of kasgot to a 1:1 coir blend. Dried output/sales: 5.97/4.78 kg at assumed Rp28,000/kg. Media output/sales: 101.56/76.17 kg. Household ownership and all processing costs remain.

8.2 Service capacity and recurring coordinator costs

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Shortest-month commercial capacity is 33.33 × 85% × 28 × 80% = 634.67 kg. One mature home can reserve three 200 kg plans, and half-capacity commissioning permits one. Plans are prepaid. Accepted excess requires an active agreement and has a one-month cash lag. The average month is 365.25/12 days.

Table A8.6. Recurring coordinator cost drivers.

Cost	Base basis
Collection and delivery	Daily visits × Rp8,000/customer stop + Rp350/commercial kg. Route labor, vehicle, fuel and home delivery included.
Specialist service	Rp250,000/month + Rp25,000/home + Rp75/accepted kg.
Coordination hours	Maximum of 16 and 8 + 2 × homes + 1.5 × accounts per month.
Administration hours	Maximum of 8 and 4 + 0.5 × homes + 0.25 × accounts per month.
Paid hours basis	Rp6,000,000/160 hours. Model loaded compensation, with route and specialist labor separately costed.
Technology and professional services	Rp350,000 technology + Rp500,000 professional/month. The latter allocates Rp100,000 to incident/representation administration and Rp400,000 to remaining tasks. Setup remains Rp25,000,000. Add SIRKULA-paid JKK/JKM Rp28,800 and qualified relief Rp100,000 per active home/month at 2027 planning prices.
New customer and home	Rp100,000/new account. Rp300,000/new home training.
Participant transfers	Rp3,700/commercial kg to home. Rp400/resident kg to contributor.

2027 planning basis. Recurring prices and costs escalate at 4% annually. The additional SIRKULA protection allowance is Rp128,800 per active

home/month before escalation. Existing PPE/hygiene and incident-administration budgets are allocated once. Setup, capital and timing remain separate.

Table A8.7. Outsourced and indicative owned-support full costs.

Active homes	Outsource Rp m/month	Own full cost Rp m/month
1	0.34	4.14
5	0.70	4.16
10	1.15	4.19
19	1.95	4.23
25	2.49	4.26
31	3.03	4.29

2027 Rp million/month. Owned support includes staff, occupancy, utilities and capital recovery.

Owned support capacity is 125 one-rack equivalents, or 31 four-rack homes. The approximately 46-home support-cost crossover exceeds capacity. Operator protection is a separate coordinator obligation and does not establish a new support-facility crossover. The base forecast schedules no hub.

8.3 Ten year production cash and funding

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Table A8.8. Demand-constrained production by year.

Year	Closing homes	Feed tonnes	Fresh sold tonnes	Kasgot sold tonnes
2027	1	9.9	0.95	1.35
2028	2	20.3	2.00	2.98
2029	4	40.5	3.98	5.95
2030	6	61.2	6.06	9.20
2031	8	81.9	8.14	12.44
2032	10	102.8	10.25	15.74

Year	Closing homes	Feed tonnes	Fresh sold tonnes	Kasgot sold tonnes
2033	13	133.1	13.25	20.29
2034	15	154.2	15.42	23.81
2035	17	174.9	17.50	27.06
2036	19	196.2	19.63	30.39

Closing homes are year-end counts, not annual average productive homes. Tonnes include commissioning and household addition timing.

Table A8.9. Annual SIRKULA operating and cash account.

Year	Revenue	Operating costs	EBITDA	Net cash
2027	58.29	69.62	−11.33	−58.52
2028	124.43	123.29	1.14	−22.43
2029	258.85	234.34	24.51	−28.93
2030	413.03	360.46	52.57	−7.85
2031	573.14	492.36	80.78	10.98
2032	746.20	635.15	111.05	32.45
2033	1,010.75	852.54	158.21	40.25
2034	1,215.78	1,020.38	195.40	93.73
2035	1,431.71	1,198.34	233.36	120.93
2036	1,674.79	1,397.04	277.74	154.52

Rp million. Net cash includes payment timing, cash allowances, household equipment support and investment. It differs from EBITDA.

Table A8.10. Annual funding peaks and closing balances.

Year	Peak funding	Closing cash	Receivables	Home renewal reserves
2027	82.84	−58.52	0.59	3.56
2028	123.95	−80.95	1.23	10.98

Year	Peak funding	Closing cash	Receivables	Home renewal reserves
2029	204.99	−109.88	2.55	26.40
2030	271.85	−117.73	2.81	50.46
2031	321.34	−106.75	4.30	83.82
2032	353.96	−74.30	5.91	127.19
2033	416.34	−34.05	7.07	185.83
2034	404.72	59.68	8.90	256.19
2035	369.39	180.61	10.87	306.28
2036	311.74	335.12	11.51	368.52

Rp million. Peak funding is the largest monthly shortfall to the required operating reserve within each year. Cash, receivables and household earmarked reserves are closing balances. They are different accounts.

Table A8.11. Financing requirements and return measures.

Measure	Result and definition
Ten-year NPV	Rp64,200,142 after funded protections. Monthly unfinanced SIRKULA cash at 12% annual discount. Grant receipts excluded.
Peak external funding	Rp416,335,754 maximum gap to four months of operating costs. Includes protection costs and conditional equipment outlays.
First 24-month funding	Rp123,950,910. Committed equity, grants and debt remain zero.
Operating cost coverage	2036 revenue covers 119.88% of operating costs. 2027 coverage is 83.73%. Grants are financing, not operating revenue.
Cash payback	Month 92. Cumulative unfinanced cash stays nonnegative through month 120, conditional on forecast assumptions.
Final receivables	Rp11,508,833. No terminal recovery or resale in NPV.

Entity NPV discounts monthly unfinanced SIRKULA cash at 12% annually. The 22% positive-EBITDA allowance is cash contingency, not statutory tax. Recurring protection costs are included. Carbon receipts are zero and formal development is deferred. Required capital is distinct from committed finance.

Equipment support and operator contributions are capital funding, not revenue. Consolidation cancels internal handling and sales fees. Revised pre-allowance network cash NPV is Rp1,764,743,383, including equipment and protection costs but excluding imputed owner labor and space. It differs from entity NPV.

8.4 Operating alternatives and household headroom

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Table A8.12. Operating alternatives and downside comparisons.

Case and capture basis	Home income Rp m	NPV Rp m	Funding Rp m
Base daily service	2.11	64.20	416.34
Downside	0.44	−662.12	1,479.07
Combined optimistic case	4.13	584.48	111.68
Rejected flat Rp5000/kg	2.11	−945.47	2,300.48
Thirteen collections / Rp1.35m	2.11	46.53	399.46
Twenty-six collections / Rp1.45m	2.11	39.79	425.85
Capacity plan paid one month late	2.11	−8.38	494.66
Reference carbon all-in (historical, before protections)	2.11	77.67	422.29
Hold one home	2.11	−116.28	201.96
Later rollout delayed 12 months	2.11	−0.68	441.00
Source and buyer demand 80%	2.11	21.76	367.76
Delivered volume 80%. Plans reserved	1.59	202.05	293.38
Preparation machine rate 50 kg/hour	2.11	64.20	416.34
Home task time 25% longer	2.11	64.20	416.34
Owned support from five homes (indicative)	2.11	−59.08	574.85

Case and capture basis	Home income Rp m	NPV Rp m	Funding Rp m
Rejected previous one-rack design (historical capture)	0.17	−119.87	577.70

Mature household income is in 2027 Rp million/month. Entity NPV and funding cover 120 months. Fixed deterministic comparison captures include protection costs. The inherited carbon all-in and earlier one-rack results remain labeled historical pre-protection captures. The BPJS calculation base remains a Rp2.2 million planning declaration in these cases. Confirm each operator's total declared income and required coverage before intake, then rebudget the contribution input.

At unchanged demand, the revised zero-NPV monthly plan is Rp1,461,890, leaving Rp38,110 below the proposed Rp1.5 million price. This is a narrow planning margin, not evidence of willingness to pay. Delivered volume and task time remain separate household risks.

Table A8.13. Household operating headroom and local optimization tests.

Test or limit	Base or example result	Decision
Delivered-volume floor	87.37% of planned intake. 753.46 kg/month at the base commercial mix.	Verify actual eligible deliveries before committing another home.
Economic work ceiling	69.42 hours/month at current income and workspace value. 7.80 hours above the 61.62-hour work budget.	The nonnegative economic-return test binds before the 75-hour side-job limit.
Five minutes less work per day	2.54 hours/month released. Rp63,411 greater economic residual. Cash income unchanged.	Adopt only after timed cycles demonstrate the saving.

Test or limit	Base or example result	Decision
Rp1,000 less per daily route stop	Rp91,313/month platform saving per mature home with three accounts.	Obtain a complete route quote with unchanged daily service. A Rp91,313 saving covers about 71% of the additional Rp128,800/home protection budget.
Availability-payment comparison	Rp600,000/month plus Rp2,830.33 per commercial kg preserves full-fill handling pay. At 80% fill it adds Rp120,000 household cash relative to the current rate.	Optional funded comparison only. At 80% fill the platform pays Rp120,000 more per home, in addition to funded protections. No retainer or wage guarantee is adopted in the forecast.

Availability payments reallocate risk without creating consolidated revenue. Before adopting revised terms, recalculate full cash and uncertainty effects. Two relief hours are funded standby. Further help needs approval or reduced scope. Timed records must verify any claimed owner-work saving.

Appendix 9 Historical scenario analysis

9.1 Historical scenario design and provenance

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Historical capture contains 10,000 ten-year draws, seed 20260919 and recorded NumPy PCG64 provenance. Judgmental triangular inputs share market and biology factors. Captures precede commercial-billing, capacity and protection-cost revisions and have not been rerun.

Table A9.1. Declared triangular scenario inputs.

Variable	Low	Mode	High
Plan Rp/month	Rp1,125,000	Rp1,500,000	Rp1,750,000
Fresh price Rp/kg	Rp3,000	Rp4,000	Rp5,500
Kasgot Rp/kg	Rp800	Rp1,500	Rp2,200
Fresh sell-through	0.55	0.90	1.00
Kasgot sell-through	0.35	0.80	1.00
Larval wet yield	0.09	0.12	0.16
Utilization	0.65	0.85	0.95
Larvae acceptance	0.80	0.95	1.00
Kasgot acceptance	0.75	0.90	1.00
Residue dry-matter recovery	0.75	0.90	0.95
Commercial share	0.65	0.80	0.90
Source and buyer demand factor	0.65	1.00	1.20
Operating cost factor	0.90	1.00	1.25
Capital cost factor	0.90	1.00	1.30
Equipment life months	72	96	108
Chopping kg/hour	50.00	75.00	100.00
Owner task-time factor	0.90	1.00	1.25

Variable	Low	Mode	High
Delivered intake share	0.70	1.00	1.00

Low, mode and high are planning ranges. Prices are 2027 rupiah. Quality, yield and sale quantities use shares. Multipliers scale the baseline.

Table A9.2. Stored scenario dependencies and timing.

Relationship	Declared construction
Market and product sales	Shared normal factor loads on prices and sell-through. Commercial mix shares a smaller loading.
Biology	Common factor links yield, utilization, acceptance and residue recovery.
Price and volume	Demand combines market conditions, an adverse service-price-specific effect and residual variation.
Costs capital and work	Linked operating/capital factors and uncertain equipment life. Owner task-time stress is applied explicitly.
Delivery and payment	Eligible delivered fill varies while reserved plans remain fixed. In the stored trials, capacity-plan payment lag and rollout delay move together. Their separate rows therefore do not identify independent effects.

Captured frequencies describe the original assumption-driven scenarios. They are neither fitted local probabilities nor live updated-rule results. New rules, costs or ranges require matching recalculation before revised frequencies are claimed.

9.2 Captured distributions association and stability

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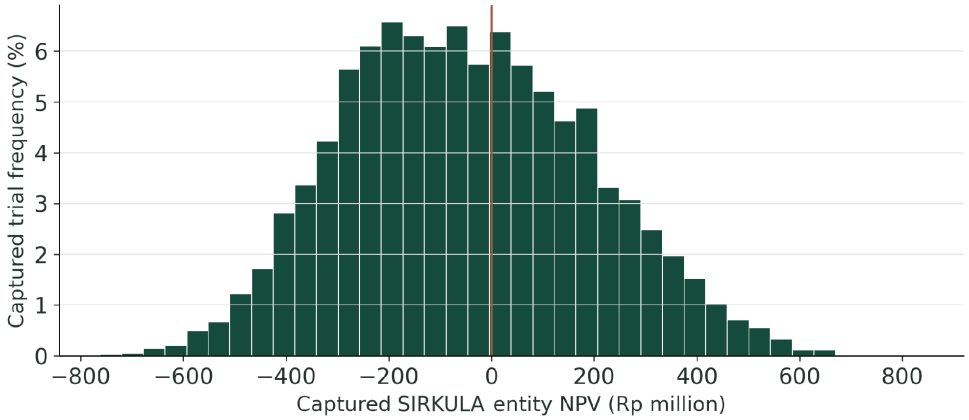


Figure A9.1. Captured SIRKULA NPV across declared scenarios. Source: Stored 10,000-trial financial scenario analysis.

Table A9.3. Captured outcome percentiles.

Outcome	P10	Median	P90
Home income Rp/month	Rp1,112,464	Rp1,626,918	Rp2,178,130
Home income Rp/hour	Rp20,669	Rp27,731	Rp34,899
Home hours/month	50.8	58.2	67.0
Home economic residual Rp/month	−Rp613,925	−Rp215,502	Rp226,865
Entity NPV Rp million	−348.72	−59.52	273.33
Peak funding Rp million	263.57	484.00	964.38
First 24-month funding Rp million	91.74	125.99	155.42

Table A9.4. Captured shares meeting the stated objectives.

Objective	Trial share
SIRKULA NPV above zero	41.47%
Home income at least Rp1m/month	94.86%
Income at least Rp25,000/hour	68.51%

Objective	Trial share
Normal owner work at most 75 h/month	99.44%
Platform NPV plus income and hourly goals	37.13%
Nonnegative home economic residual	26.43%
All five platform and side-job goals	18.72%

Captured objectives combine entity NPV, household income, hourly earnings, work and economic residual under the original scenario basis. The current protection-cost forecast and separate twelve-rack case are outside these captured trials.

Table A9.5. Rank associations in the captured trials.

Driver	NPV ρ	Home income ρ
Service plan price	0.87	0.35
Common market factor	0.67	0.53
Capacity plan payment lag	−0.62	−0.46
Rollout delay months	−0.62	−0.46
Excess service payment lag	−0.46	−0.38
Commercial share	0.45	0.79
Kasgot sell-through	0.43	0.37
Fresh sell-through	0.42	0.41

ρ is Spearman rank association in the coupled historical trial design. Capacity-plan lag and rollout delay move together, so their identical associations do not identify independent causal effects.

Table A9.6. First 5,000 versus all 10,000 captured draws.

Outcome	Maximum quantile shift / P90–P10 width	Pass
SIRKULA NPV	0.58%	Yes
Peak funding	0.56%	Yes

Outcome	Maximum quantile shift / P90–P10 width	Pass
Household income	0.59%	Yes
Income per hour	0.78%	Yes
Owner hours	1.08%	Yes
Economic residual	0.83%	Yes

Maximum quantile shift is divided by the P90–P10 width. The recorded threshold is 5%. Positive-NPV frequency shift is 0.11 percentage points against a one-point limit, with 0.49-point frequency standard error.

Convergence measures numerical precision, not input validity. Captures do not model random exits/default. Lag and rollout delay are coupled, not independent causal effects. Operating records can inform updated assumptions (U.S. Government Accountability Office, 2020).

Appendix 10 Safeguards operational risks and product release

10.1 Funded safeguards exceptions and risk owners

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Table A10.1. Funded protection and operating controls.

Control	Planning amount and payer	Implementation requirement
Operator JKK and JKM	SIRKULA: Rp22,000 JKK plus Rp6,800 JKM per active home/month.	JKK planning base is the declared Rp2.2 million income band. Confirm registration, declaration, eligibility and official contribution before intake. Do not assume a temporary discount.
Qualified relief	SIRKULA: two hours/home/month at Rp50,000/hour = Rp100,000.	Standby budget, not an automatic deduction from owner hours. Record deployment, competency, task substitution and the actual quote/worker classification. Additional hours need a funded approval or reduced scope.
Household PPE and hygiene	Rp50,000 within the existing Rp200,000 household utilities/hygiene allowance.	Record suitable protective supplies, routine checks and replenishment. Do not add the same cost a second time.
Incident and representation administration	Rp100,000 within the existing Rp500,000/month SIRKULA professional-services budget. Existing paid coordination/administration hours also apply.	Provide incident logging, independent contested-charge review and orderly exit support. Material extra work requires a budget review.
Capacity and safe workload	Existing task, tray, stock, technical-support and funded-relief budgets.	Reschedule finishing, activate qualified paid relief or reduce starts. Never resolve underfill by accepting unsafe loads.

Control	Planning amount and payer	Implementation requirement
Operating reserve and paid collection backup	Existing costed route/support obligations and model cash reserve.	Use named providers, agreed diversion and payment records. Reserve cash is funding capacity, not new revenue.

Source: BPJS Ketenagakerjaan (2026) and authors' revised contribution and protection budgets.

Incremental SIRKULA protection allowance is Rp128,800 per active home/month at 2027 planning prices. Household available income remains unaffected because SIRKULA pays that allowance. Premiums and relief prices require confirmation before launch. No additional injury-compensation or business-loss guarantee is modeled.

Table A10.2. Operating signals and response responsibilities.

Signal	Immediate response	Owner
Contamination	Reject/segregate. Record destination and correct source instructions.	Intake operator.
Poor biology or late starters	Isolate abnormal lots. Reject deteriorated stored feed. Reduce starts and activate approved starter backup.	Technical lead.
Odor, pests or runoff	Contain, clean and suspend intake pending inspection.	Home and technical lead.
Rejected or aging stock. Missed visits	Limit production to demand. Use qualified outlets or costed collection backup.	Sales/route coordinators.
Arrears, deficit or overload	Review income, hours and unpaid balances. Reschedule tasks, limit intake or suspend expansion.	Finance/operations with operator representative.

Contain incidents, record a responsible owner and obtain restart approval. Costed route/support backup and the funded two-hour relief allowance address bounded interruptions. Additional assistance requires approved money or reduced intake and starts.

Table A10.3. Ordinal inherent and planned residual risk.

ID	Risk	Inherent L × I	Planned residual L × I
R1	Paid-service renewal	4 × 5 = 20	3 × 5 = 15
R2	Grant and funding gap	4 × 5 = 20	3 × 5 = 15
R3	Underfilled eligible intake	4 × 4 = 16	3 × 4 = 12
R4	Workload and injury exposure	3 × 5 = 15	2 × 4 = 8
R5	Contaminated feed	3 × 4 = 12	2 × 4 = 8
R6	Starter or biological disruption	3 × 4 = 12	2 × 3 = 6
R7	Unsold or rejected output	4 × 3 = 12	3 × 3 = 9
R8	Arrears and settlement delay	3 × 4 = 12	2 × 4 = 8
R9	Odour, pests or runoff	3 × 4 = 12	2 × 4 = 8
R10	Record, access or governance failure	3 × 3 = 9	2 × 3 = 6

Likelihood and impact use 1–5 author judgments. Planned residual scores assume funded and effective controls. They prioritise decisions and do not represent observed probabilities or expected losses.

Table A10.4. Risk triggers, actions, owners and funding.

Risk and trigger	Action or control	Owner and funding
R1 Paid renewal Declined or unpaid renewal	Compare scope and bill, improve the route and use paid renewal before added homes.	Operations; acquisition, coordination and route budget.
R2 Funding No executed allocation or cash below reserve	Stage commitments and procure only funded homes; reconcile existing obligations.	Finance; separately secured capital and operating reserve.

Risk and trigger	Action or control	Owner and funding
R3 Underfill Delivered volume or household return below floor	Consolidate eligible contracted sources within capacity; revise routes/starts and stop added homes when returns fail.	Operations; paid coordination, route and contracted capacity.
R4 Work and injury Unsafe work or excessive hours	Isolate unsafe work, reschedule finishing, deploy qualified funded relief or reduce starts.	Technical lead; operations deploys PPE/hygiene, JKK/JKM and two hour relief budget.
R5 Unsafe feed Prohibited or unsafe intake	Reject and segregate; record destination and correct source guidance. Keep rejects out of paid accepted mass.	Pangantara with technical escalation; intake, hygiene and specialist budgets.
R6 Biology or starters Late starters or abnormal batch	Use specialist backup, isolate batches, reduce starts and obtain technical restart approval.	Technical lead; specialist, starter and disposal allowances.
R7 Non sale output Buyer rejection or aging stock	Hold nonconforming stock, revise starts to demand and use qualified costed outlets. Retain non sale mass in account.	Operations with household seller; sales support, finishing and management costs.
R8 Settlement Unpaid amount or unmatched statement	Keep receivables, limit exposure, use reserve and correct billing before unrelated netting.	Finance; reserve, paid administration and incident review allocation.
R9 Nuisance Complaint runoff or hygiene failure	Contain and clean, suspend affected intake, use agreed diversion and obtain restart approval.	Technical lead; household hygiene, specialist and diversion budgets.

Risk and trigger	Action or control	Owner and funding
R10 Records and governance Dispute access breach or lost allocation	Preserve evidence, restrict access, restore/reconcile and arrange independent review before further payments.	Finance; technology, professional services and paid administration.

Source: Authors' risk response design and revised financial cost allocations.

Sustained owner income and economic residual remain gates even when the coordinator retains prepaid service receipts. Relief beyond two funded hours requires explicit financing approval. No row creates guaranteed compensation or hidden staff.

10.2 Product lot evidence and release

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Table A10.5. Lot evidence and hold rules.

Stage	Record	Hold rule
Source, starters and trays	Source age/condition. Accepted/rejected kg. Starter lot. Tray dates and cumulative feed.	No dilution of prohibited material or tray overloading. Isolate abnormal stock.
Larvae harvest	Fresh weight, grade and product allocation.	Deduct rejects and retained biomass before offering stock.
Kasgot finishing	Maturation dates, raw/finished weights, moisture, screening and intended use.	Raw frass is not finished fertilizer. Hold nonconforming lots.
Liquid development	Defined source/formulation, processing, crop-specific evidence and testing cost.	Raw lindi has no automatic release or revenue.
Dispatch and recall	Lot, moisture basis, acceptance and downstream destination.	Hold implicated lots until technical approval. Preserve correction history.

Source age continues through delivery. Holding periods do not replace maturity/hygiene checks. Released lots retain buyer-specific evidence and destination.

Appendix 11 Impact measurement review and replication

11.1 Target contributions outcome definitions and denominators

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Table A11.1. SDG target contributions, measures and limits.

Target	Design link and measure	Interpretation and limit
8.3 Enterprise and formalisation	Household-owned productive assets, service/product records and an assisted acceptance pathway. Record approved and active enterprises, lawful registration status where applicable and real available income.	Proposed enterprise contribution. Model participation does not establish formalisation or measured job creation.
8.8 Safe work and rights	Costed PPE/hygiene, incident process, JKK/JKM support, paid relief, voluntary exit and independent complaints. Track confirmed coverage, incidents, corrective closure and total owner/relief work.	Controls support safer work. A contribution budget does not prove enrollment, complete protection or absence of injury.
9.3 Finance and market access	Conditional grants, documented asset title, buyer acceptance and direct household sales. Record actually funded grants and repeat accepted orders.	Grant and value-chain access are proposed. No sponsor, affordable-credit facility or guaranteed product offtake is assumed.
1.4 Economic resources and assets	Owned household equipment, transparent contributions, assisted applications and consistent site/time criteria. Track applications, acceptance reasons, actual grant recipients and ownership handovers.	Asset access is one relevant dimension. Space and capital constraints remain, and no universal access or poverty reduction is inferred.
12.5 Recycling and recovery	Accepted eligible organic kilograms, retained/rejected material, products and non-sale outputs. Report material allocations and destinations once.	Direct resource-recovery contribution. Accepted input is not itself a verified emissions reduction.

Target	Design link and measure	Interpretation and limit
8.5 Employment and equal-pay aim	Track real income, owner/relief hours and participation distributions, with equal access and transparent charges.	Qualified pathway only. Supplementary enterprise earnings do not establish full employment, equal pay or employee-wage equivalence.
12.3 Food-waste prevention aim	A future source-prevention comparison would need generated food waste or losses against a defined baseline, adjusted for activity.	Diversion of existing waste alone does not demonstrate reduced generation or the target's halving objective. No prevention outcome is modeled.

Source: United Nations Department of Economic and Social Affairs (n.d.-a, n.d.-b, n.d.-c, n.d.-d), with authors' project measures.

These project measures describe contributions to SDG targets rather than achievement of national indicators. Household records retain defined denominators and review stages.

Table A11.2. Household and service measures.

Measure	Definition	Evidence/interpretation
Cohort retention	Active at months 3, 6 and 12 ÷ homes commissioned in that entry cohort.	Register distinguishes trained, equipped, active, temporarily suspended, exited and untraced homes.
Income and hours	Income after direct charges and renewal saving. Divide by all owner work hours.	Monthly distributions separate receivables/cash. Diaries include finishing, selling, travel, records and peak tasks.
Economic/real return	Value labor once. Deflate nominal income using a disclosed baseline price index.	Report noncash appraisal separately. Use 2027 prices for real outcomes and the stated wage comparison only as context.

Measure	Definition	Evidence/interpretation
Payment and complaints	On-time payments ÷ payments due. Timely resolutions ÷ complaints due.	Monthly registers separate customer, handling and reward streams. Retain overdue cases.
Acceptance/sell-through	Accepted kg ÷ offered kg. Sold kg ÷ saleable kg available.	By product/moisture basis. Report stock, rejects and losses.
Service/renewal	Compliant visits ÷ scheduled visits. Renewals ÷ contracts reaching renewal.	Route/contract registers. Exclude new signups from renewal denominator.

Table A11.3. Capital, progression and coordinator measures.

Measure	Calculation/basis	Decision
Support efficiency	Nonhousehold cohort capital ÷ cohort homes active at 12 months.	Report original cohort and survivors.
Entry and renewal	Separate contribution plus working cash from full installed investment. Renewal cash ÷ next replacement cost.	Distinct affordability/payback denominators. Replacement funding.
Same-home growth	Incremental net income/hours after added space, equipment, diverted output, processing and qualification costs.	Four-rack retention or separately funded 12-rack enterprise. No automatic upgrade.
Platform and funding	Full operating costs ÷ accepted feed. Required cash less committed financing.	Route/support efficiency and external funding gap.

Report income and work distributions, cohort statuses and confirmed protection coverage. Deflate earnings with a disclosed index. Environmental accounts retain alternative treatment, accepted and rejected mass, products, transport and residues.

11.2 Review stages and replication decisions

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Table A11.4. Operating and outcome review schedule.

Stage	Record/comparison	Limit
First four paid weeks	Baseline earnings, assets and hours. Early collection, settlement and renewal check.	An early service milestone. Full residue finishing and sale may follow later.
Months 3, 6 and 12	First expansion review after three operating months, at least three grow-out cycles, two paid larvae lots and a qualified kasgot sale. Track later cohorts, real income, reserves and repeat buyers.	Include temporary suspensions and exits. Before/after association is not causation.
Expansion review	Incremental demand, routes, support and capital with/without expansion.	Models and scenario frequencies remain outputs of the declared operating ranges.

Review early service at four weeks, commissioning at three months, then retention, real earnings, reserves and repeat customers at months 6/12. Before/after association does not establish causation.

Table A11.5. Evidence required before additional commitments.

Gate	Evidence	If unmet
Household	At least Rp1m/month, Rp25,000/hour, at most 75 normal h/month and nonnegative economic residual after owner time and space.	Improve process, terms or scope. Suspend expansion when the side-job objective fails.
Paying demand	Paid renewal and repeated product orders. At least 753.46 total kg/month, including 602.77 commercial kg, at the base mix and prices.	Hold enrollment. Reduce starts.

Gate	Evidence	If unmet
Technical/support	Stable cycles, starters, nuisance controls, confirmed protection, paid staffing, QA, relief and backup.	Derate capacity or secure funded qualified support before more commitments.
Funding/governance	Executed grant allocation, operator contribution and working cash, operating reserve, clear statements and appeal.	Do not procure or commission unfunded homes. Correct funding and control gaps.

At the unchanged household prices and mix, the volume floor is 87.37%, or 753.46 kg/month including 602.77 commercial kg. Reservations alone do not provide it. Recalculate if household terms, work or costs change.

Assess twelve-rack capital, work, protection and household requirements separately. Replication repeats local source, buyer, route, site, support and capital checks. Neither projected Depok demand nor planned protection coverage transfers automatically.

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